Computer Networks

4th Semester, Academic Year 2026

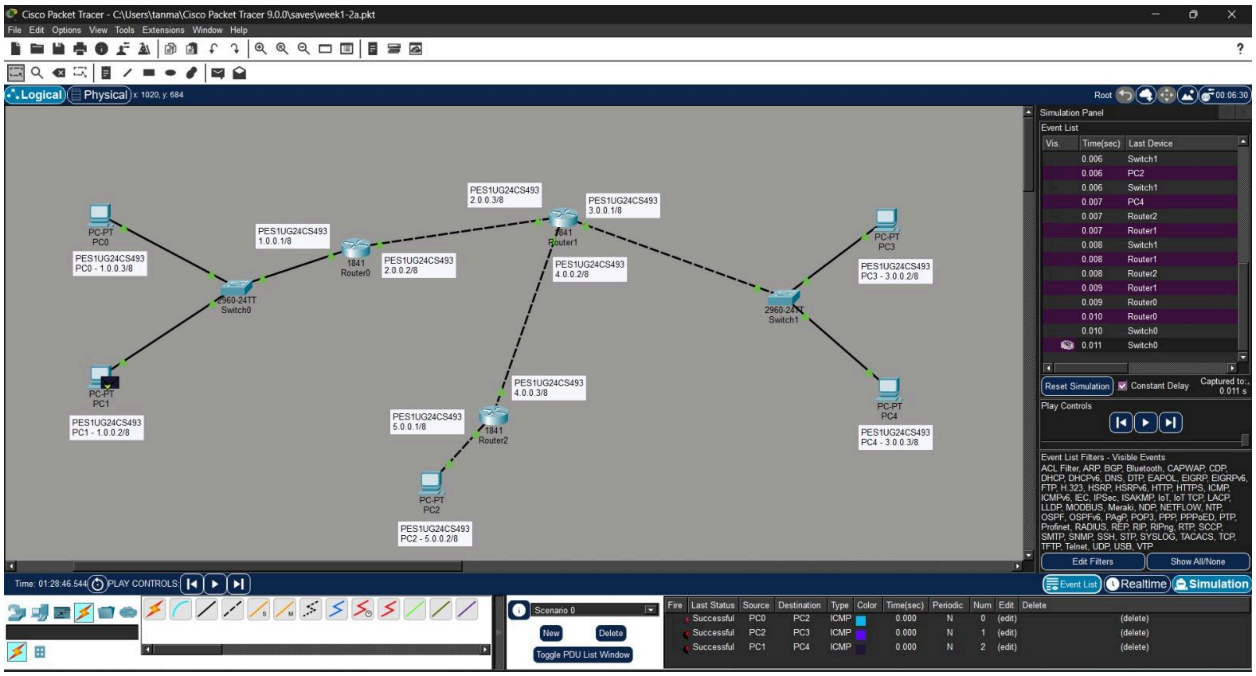
Date: 16/01/2026

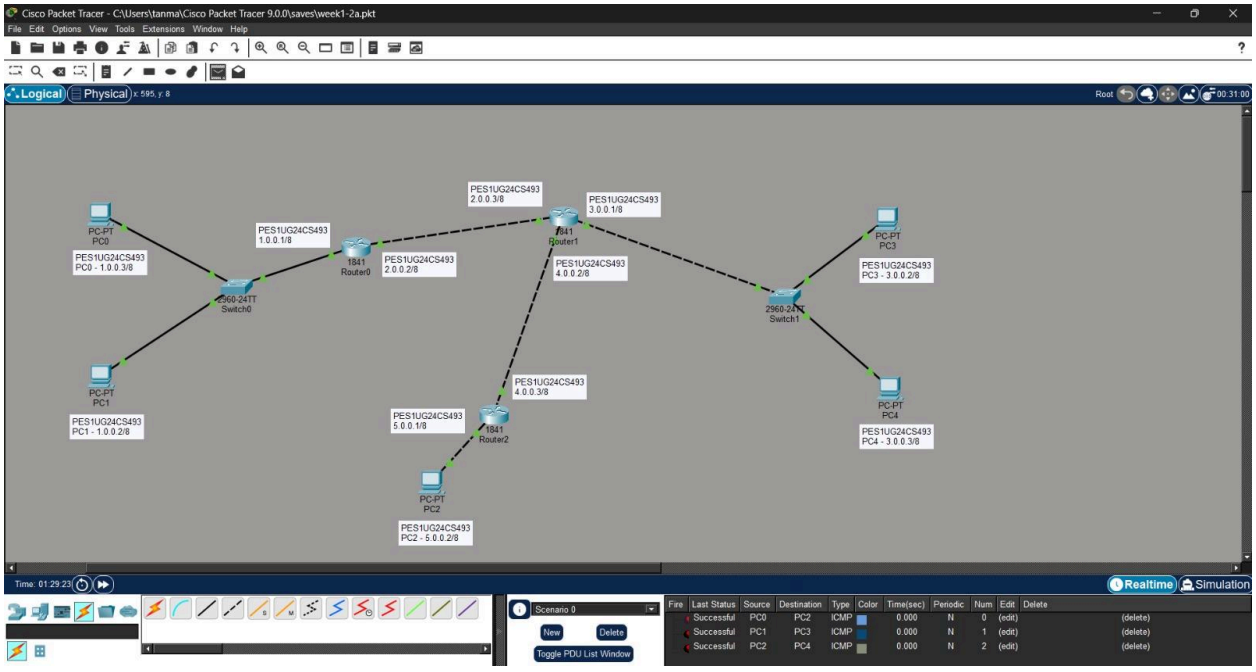
Name: Tanmayi Nagabhairava	SRN: PES1UG24CS493	Section I

Deliverables

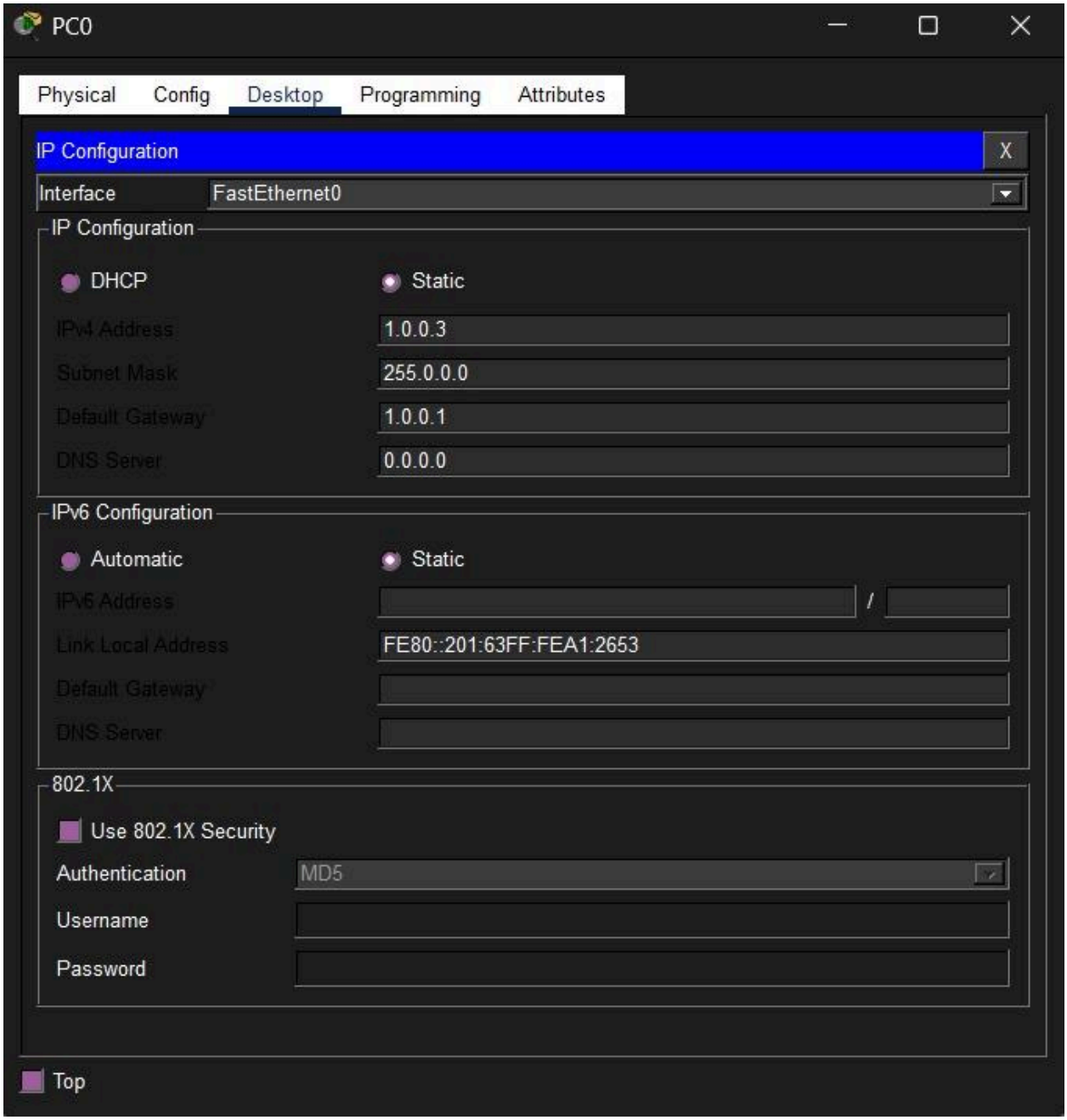
TASK 2a

Output

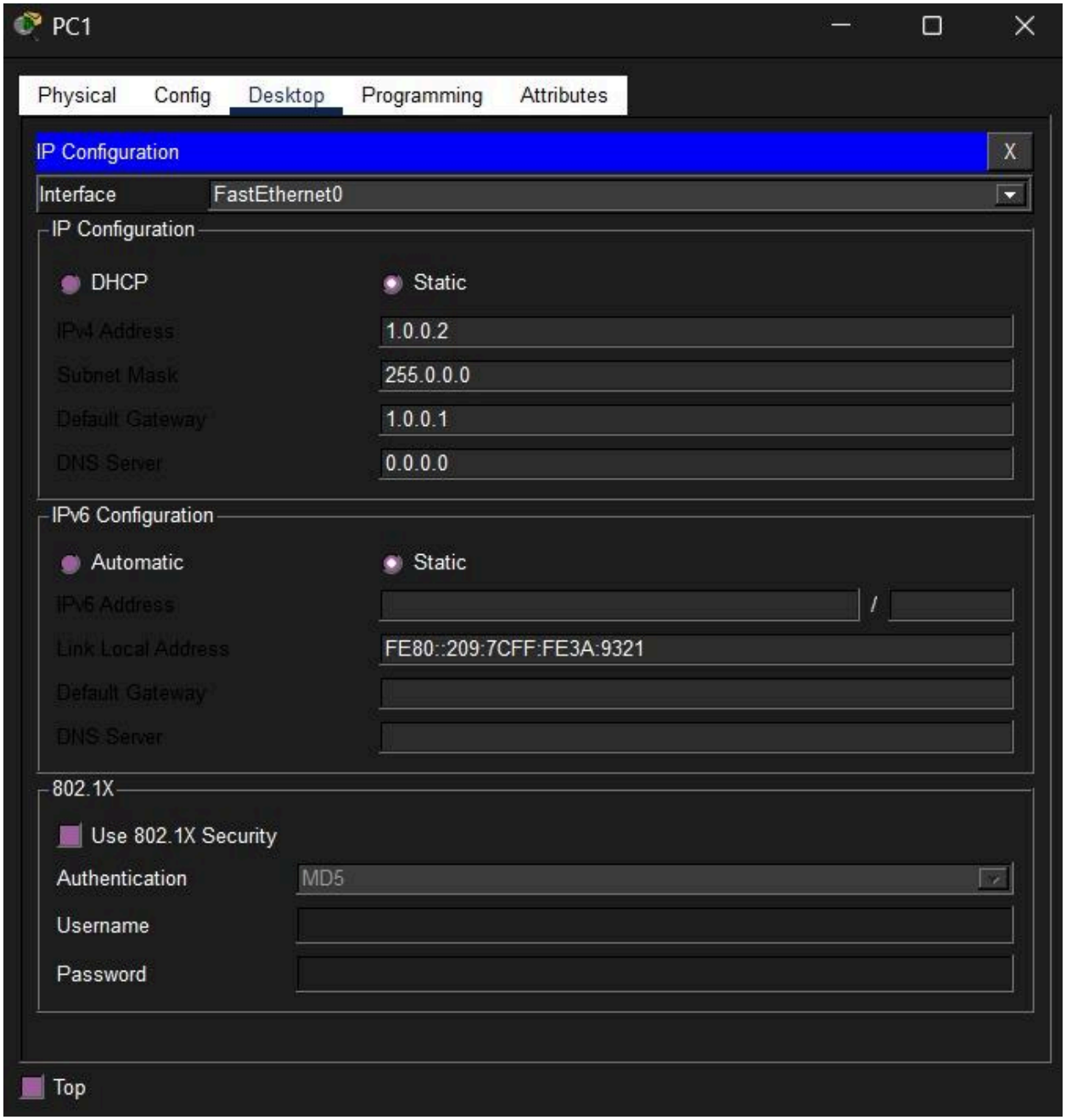



Cisco Packet Tracer - C:\Users\tanma\Cisco Packet Tracer 9.0.0\saves\week1-2a.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
PC-PT PC0
PES1UG24CS493 PC0 - 1.0.0.3/8
PES1UG24CS493 1.0.0.1/8
1841 Router0
PES1UG24CS493 2.0.0.2/8
PES1UG24CS493 2.0.0.3/8
PES1UG24CS493 3.0.0.1/8
1841 Router1
PES1UG24CS493 4.0.0.2/8
PC-PT PC3
PES1UG24CS493 PC3 - 3.0.0.2/8
2960-24TT Switch0
2960-24TT Switch1
PC-PT PC1
PES1UG24CS493 PC1 - 1.0.0.2/8
PES1UG24CS493 4.0.0.3/8
PES1UG24CS493 5.0.0.1/8
1841 Router2
PC-PT PC2
PES1UG24CS493 PC2 - 5.0.0.2/8
PC-PT PC4
PES1UG24CS493 PC4 - 3.0.0.3/8
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC0 PC2 ICMP 0.000 N 0 (edit) (delete)
Successful PC1 PC3 ICMP 0.000 N 1 (edit) (delete)
Successful PC2 PC4 ICMP 0.000 N 2 (edit) (delete)

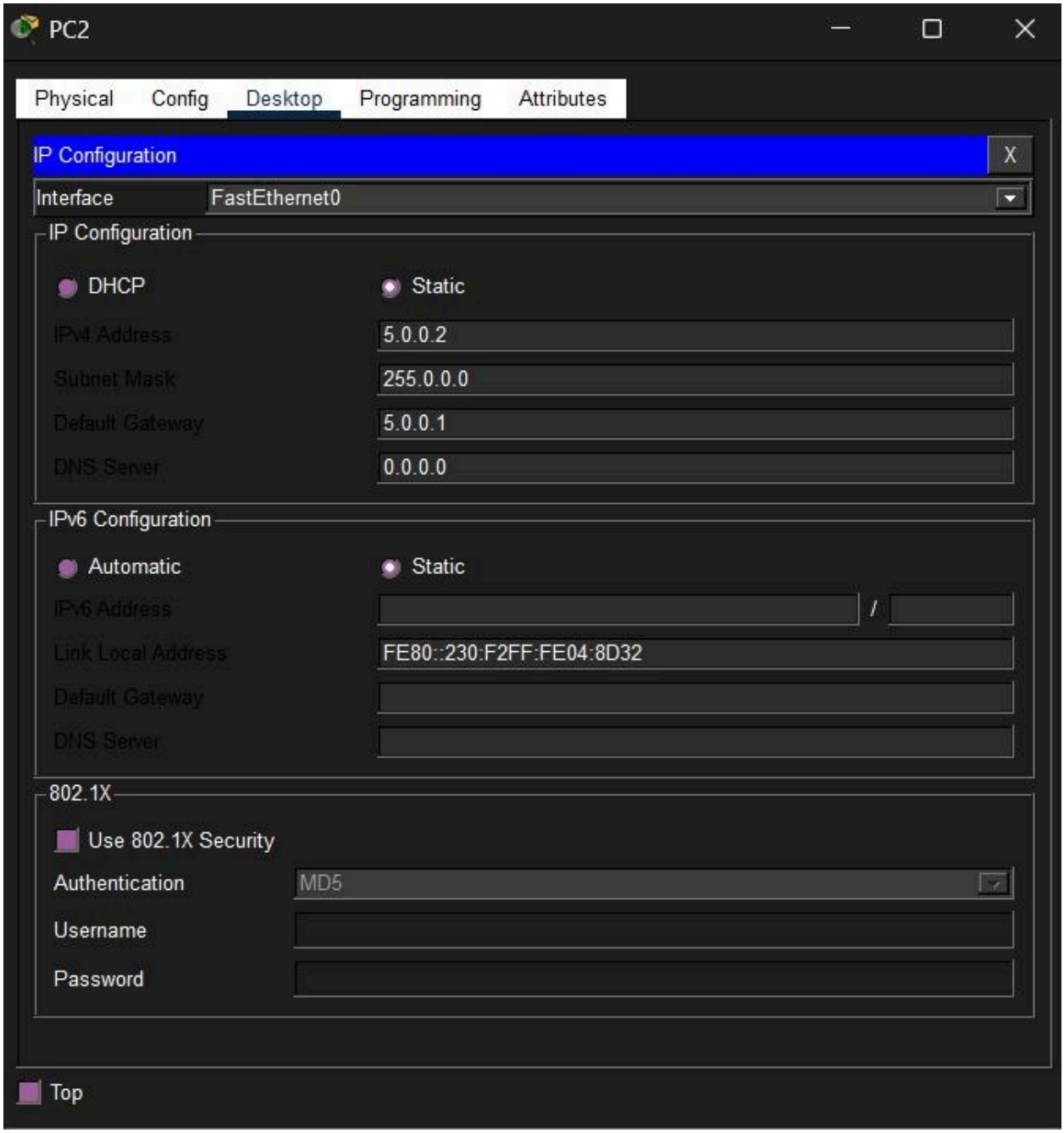
PC0


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
1.0.0.3
Subnet Mask
255.0.0.0
Default Gateway
1.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:63FF:FEA1:2653
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

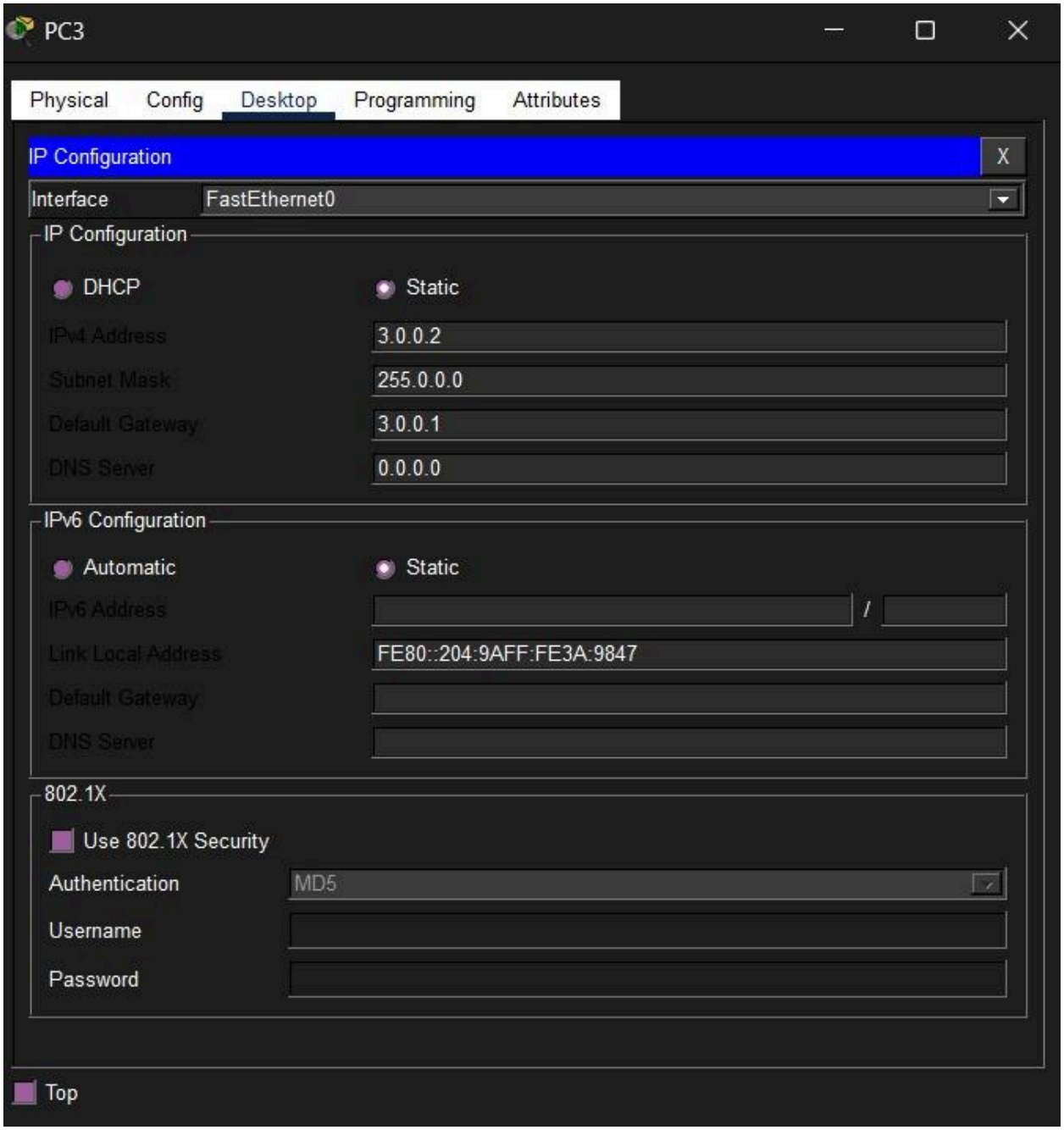
PC1


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
1.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
1.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::209:7CFF:FE3A:9321
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

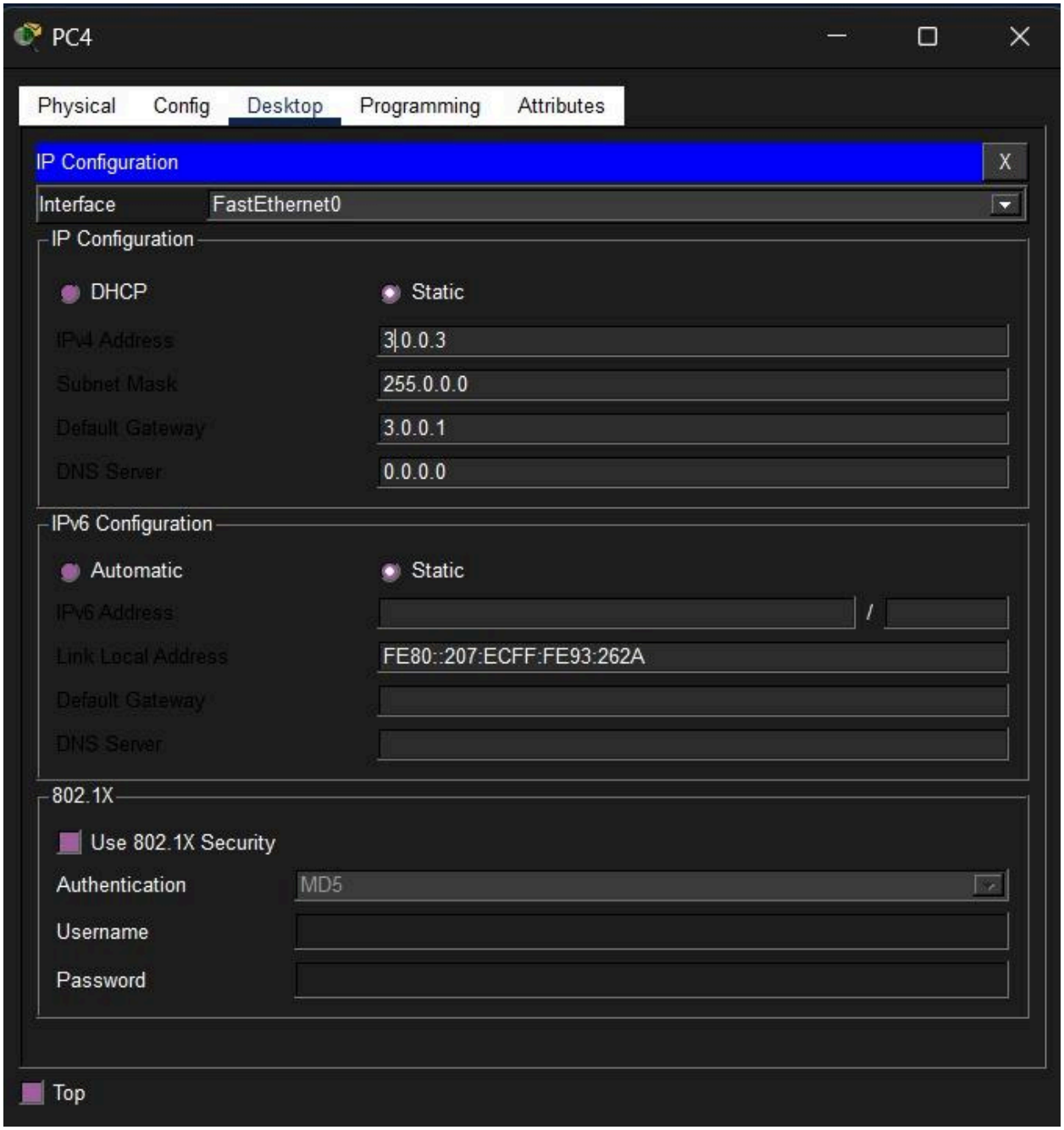
PC2


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
5.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
5.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::230:F2FF:FE04:8D32
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

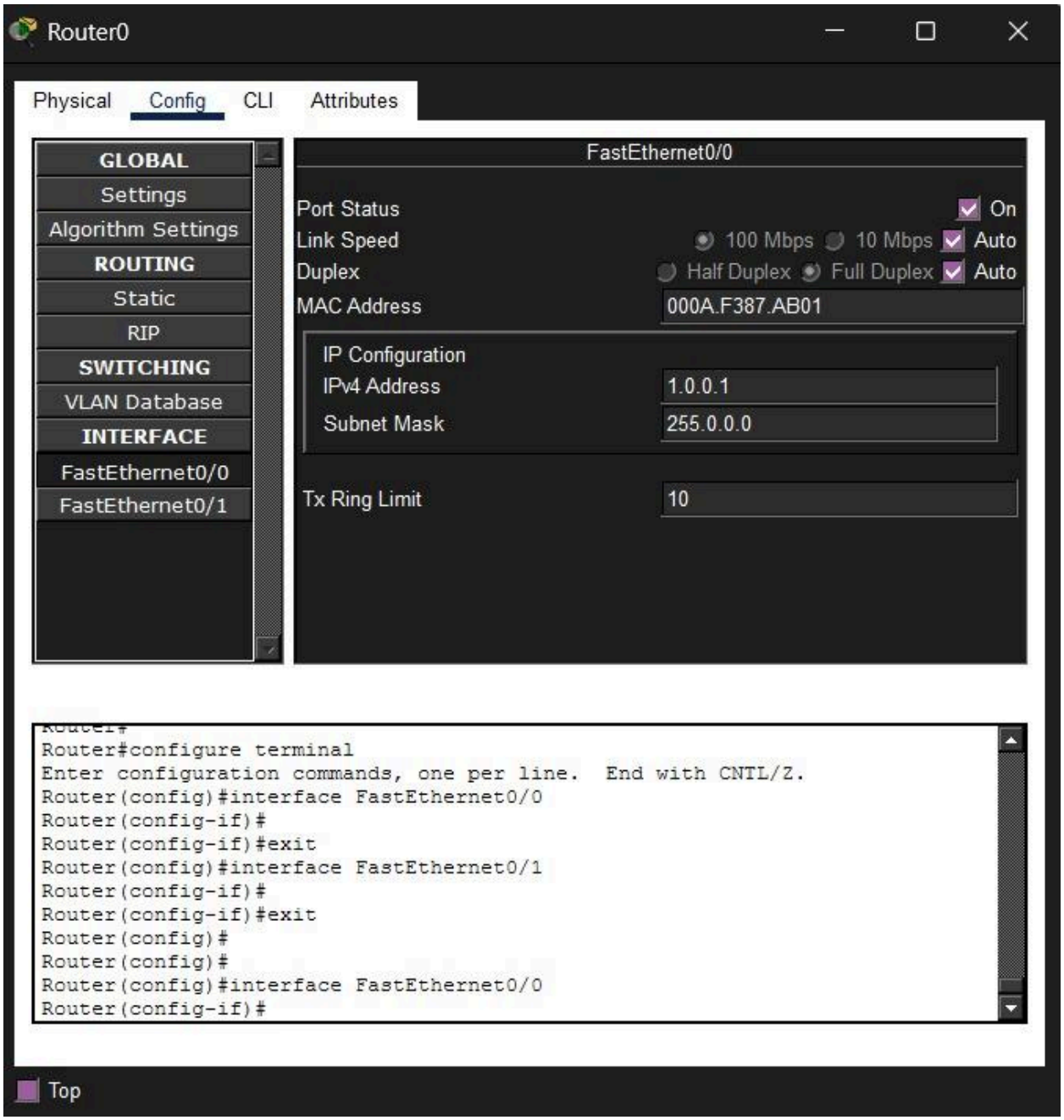
PC3

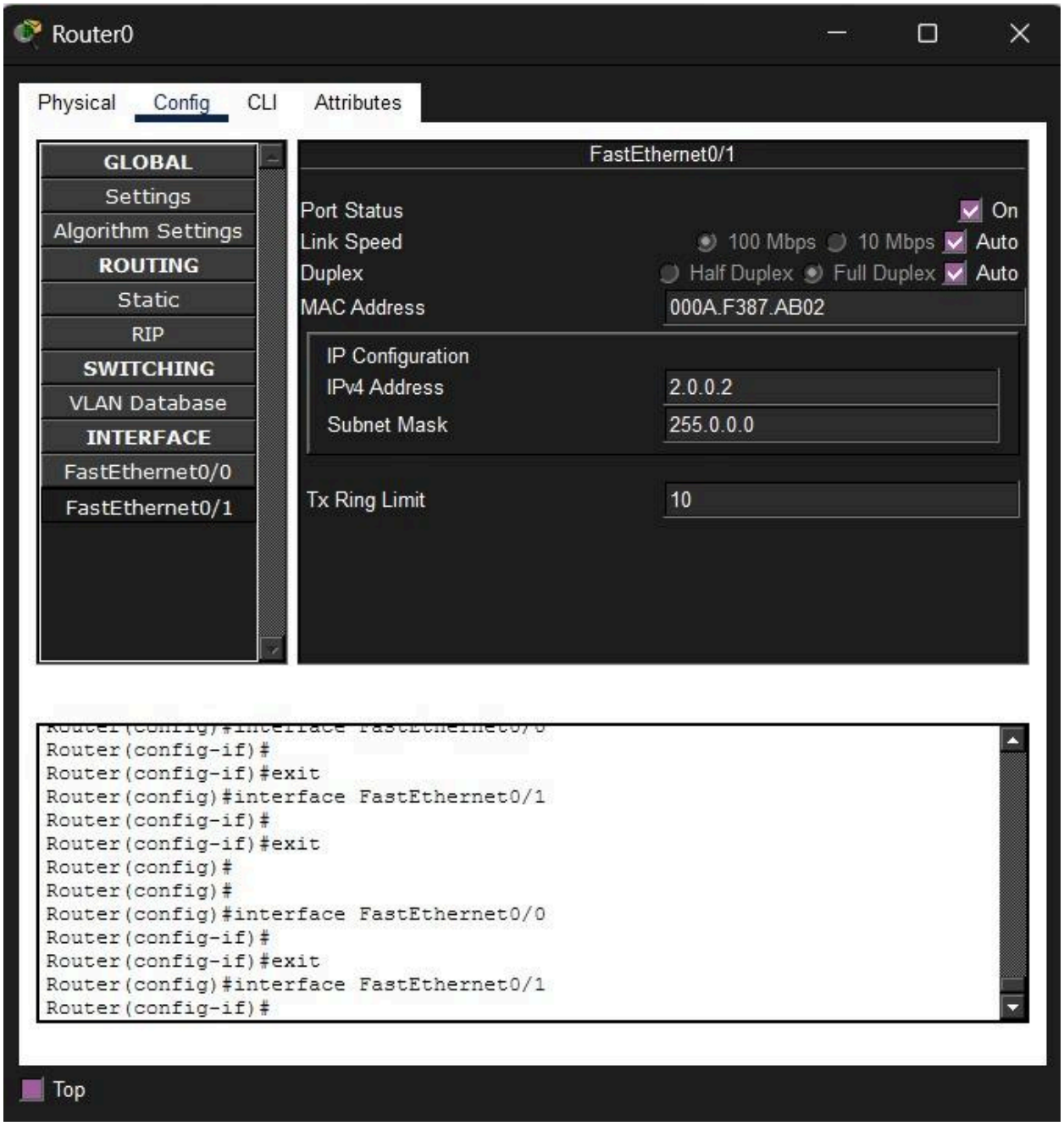

PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
3.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
3.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::204:9AFF:FE3A:9847
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

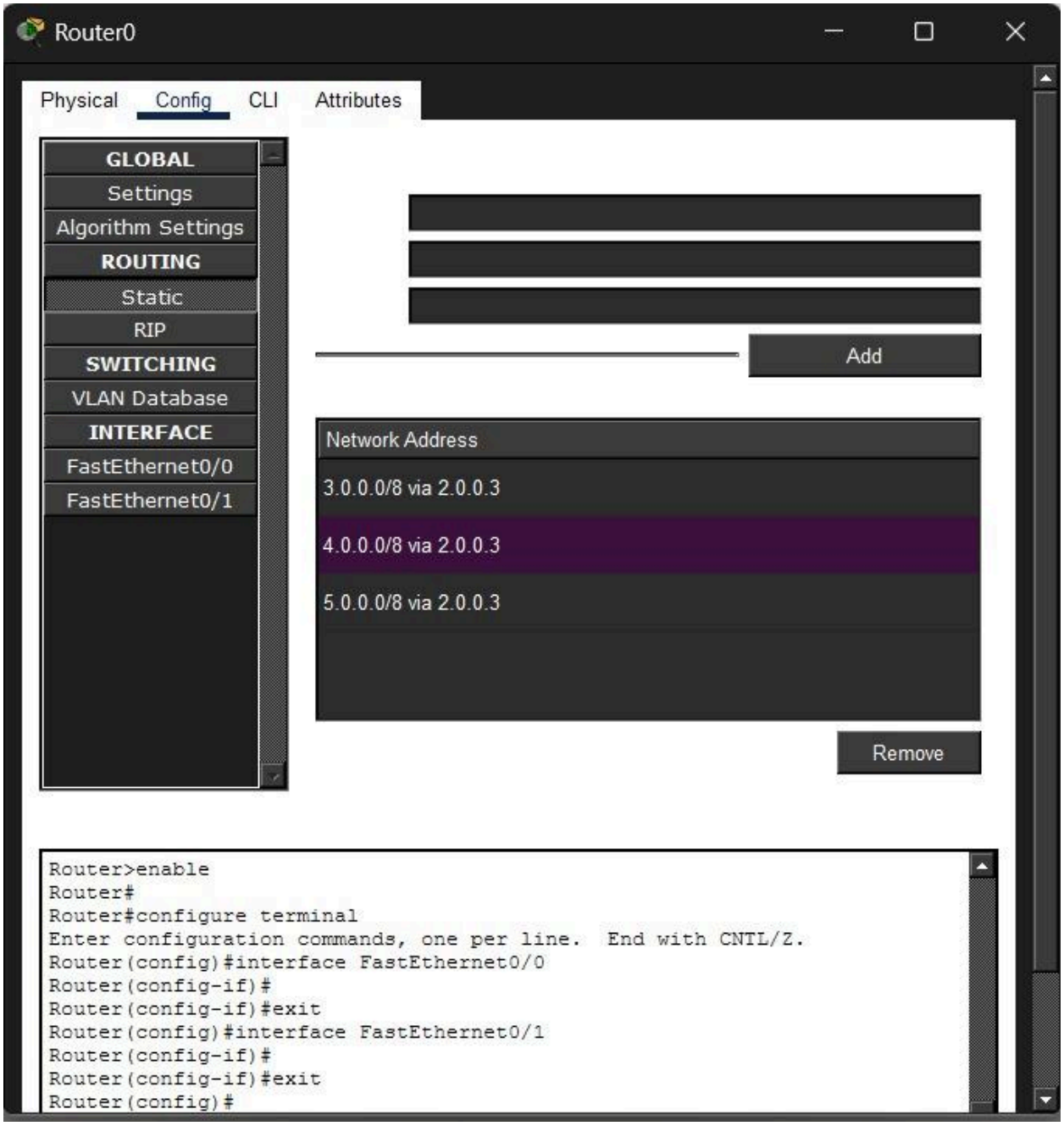
PC4


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
3.0.0.3
Subnet Mask
255.0.0.0
Default Gateway
3.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::207:ECFF:FE93:262A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

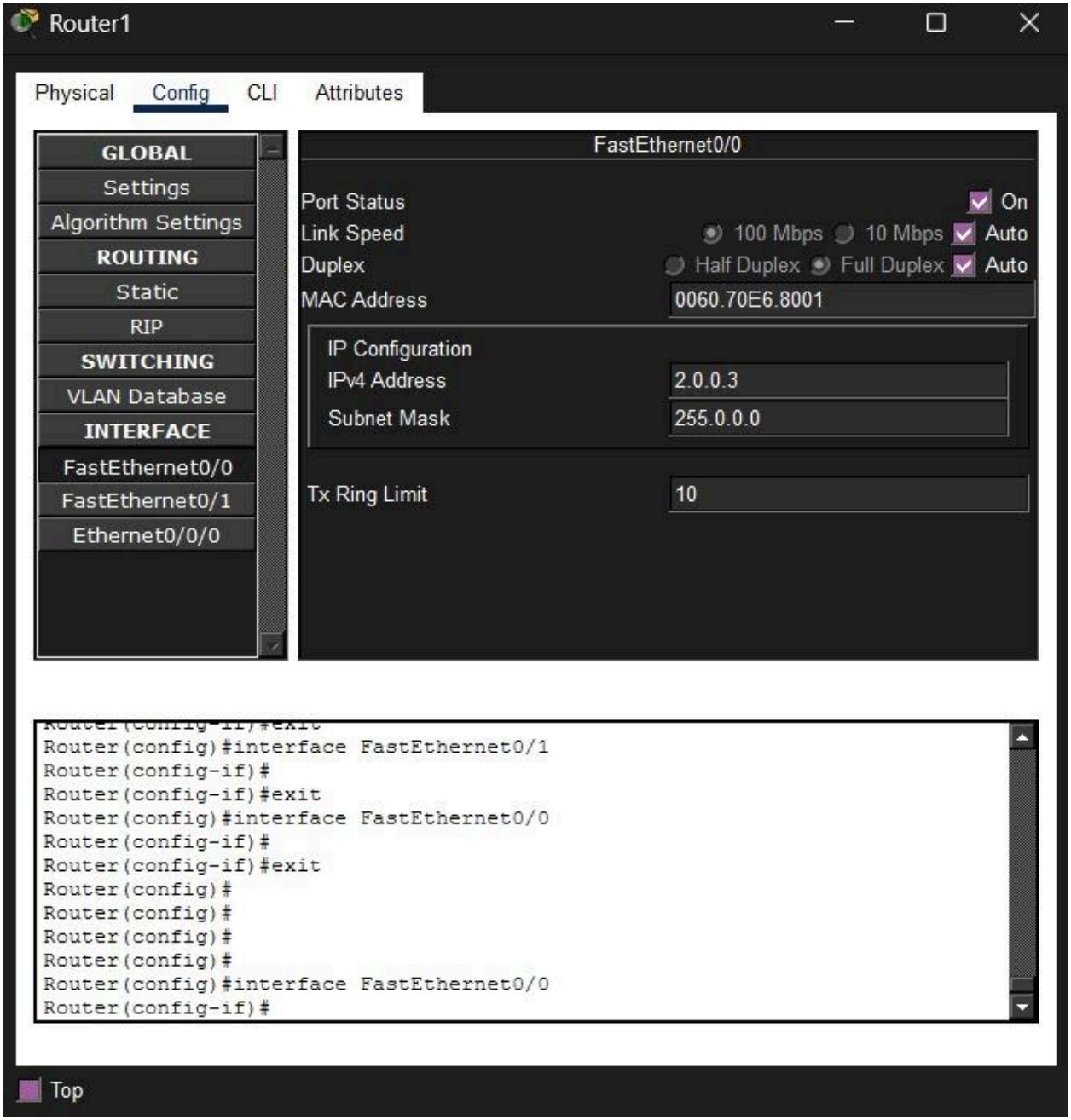
ROUTER0

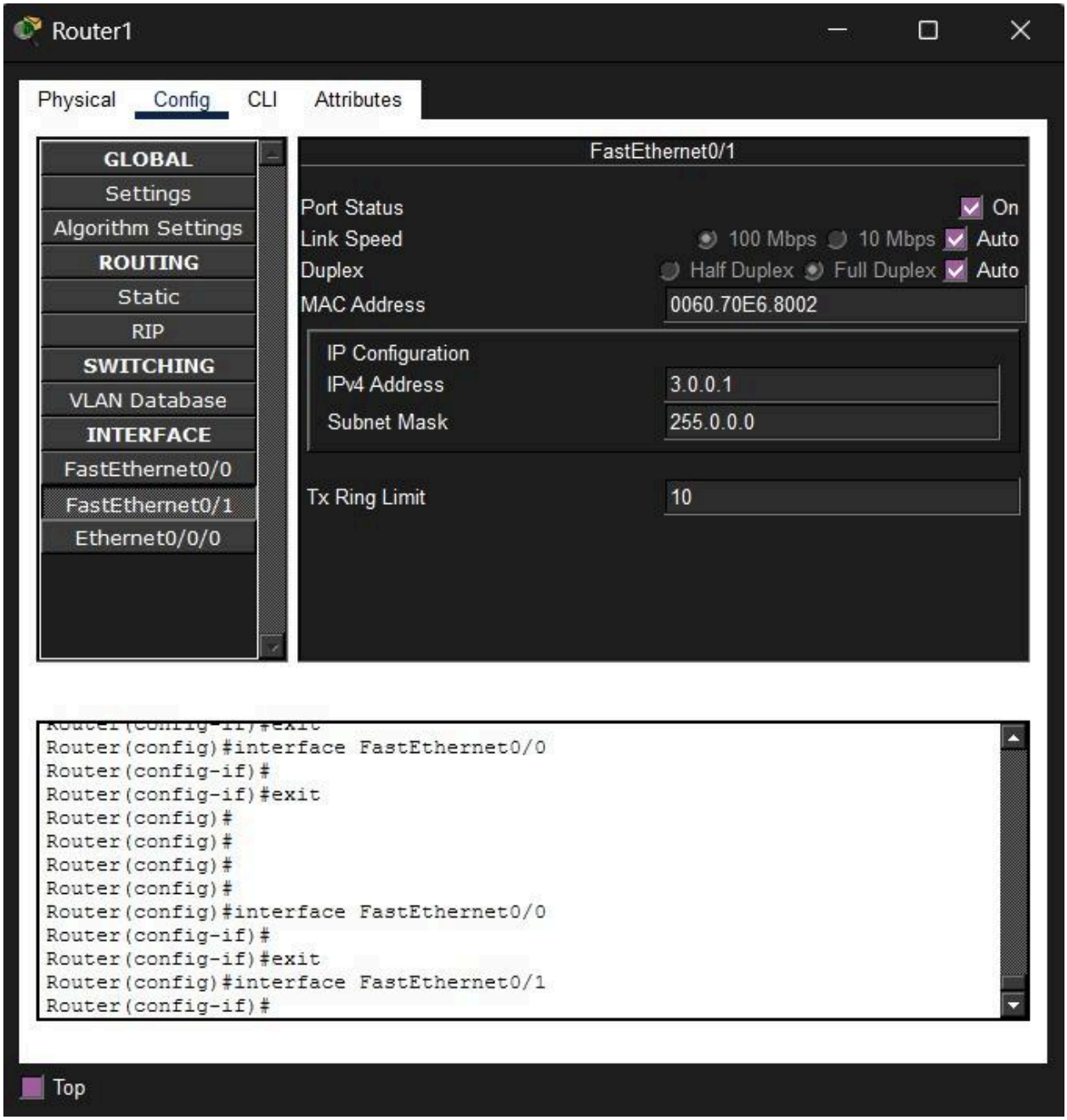

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
000A.F387.AB01
IP Configuration
IPv4 Address
1.0.0.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Top

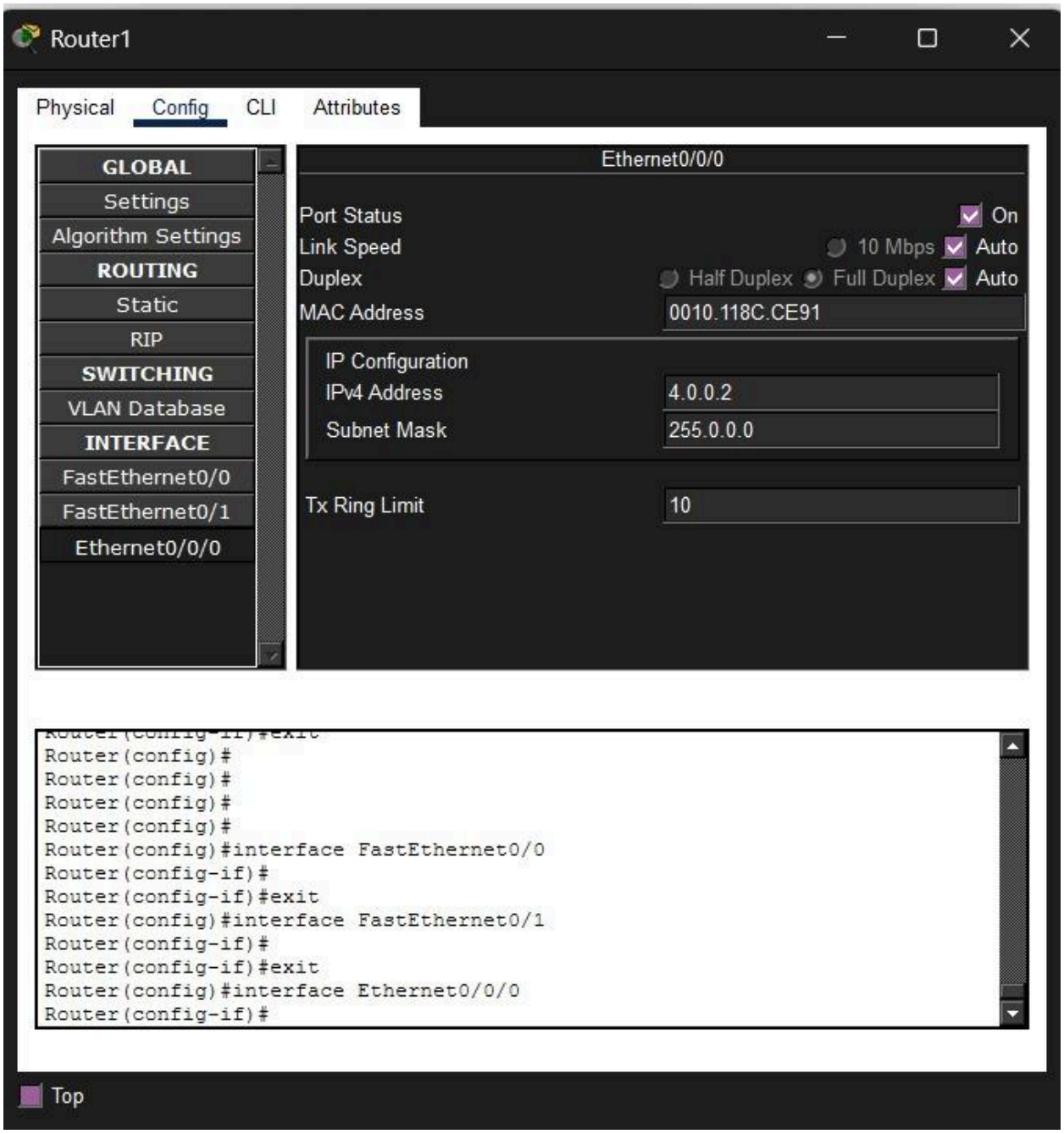

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/1
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
000A.F387.AB02
IP Configuration
IPv4 Address
2.0.0.2
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Top

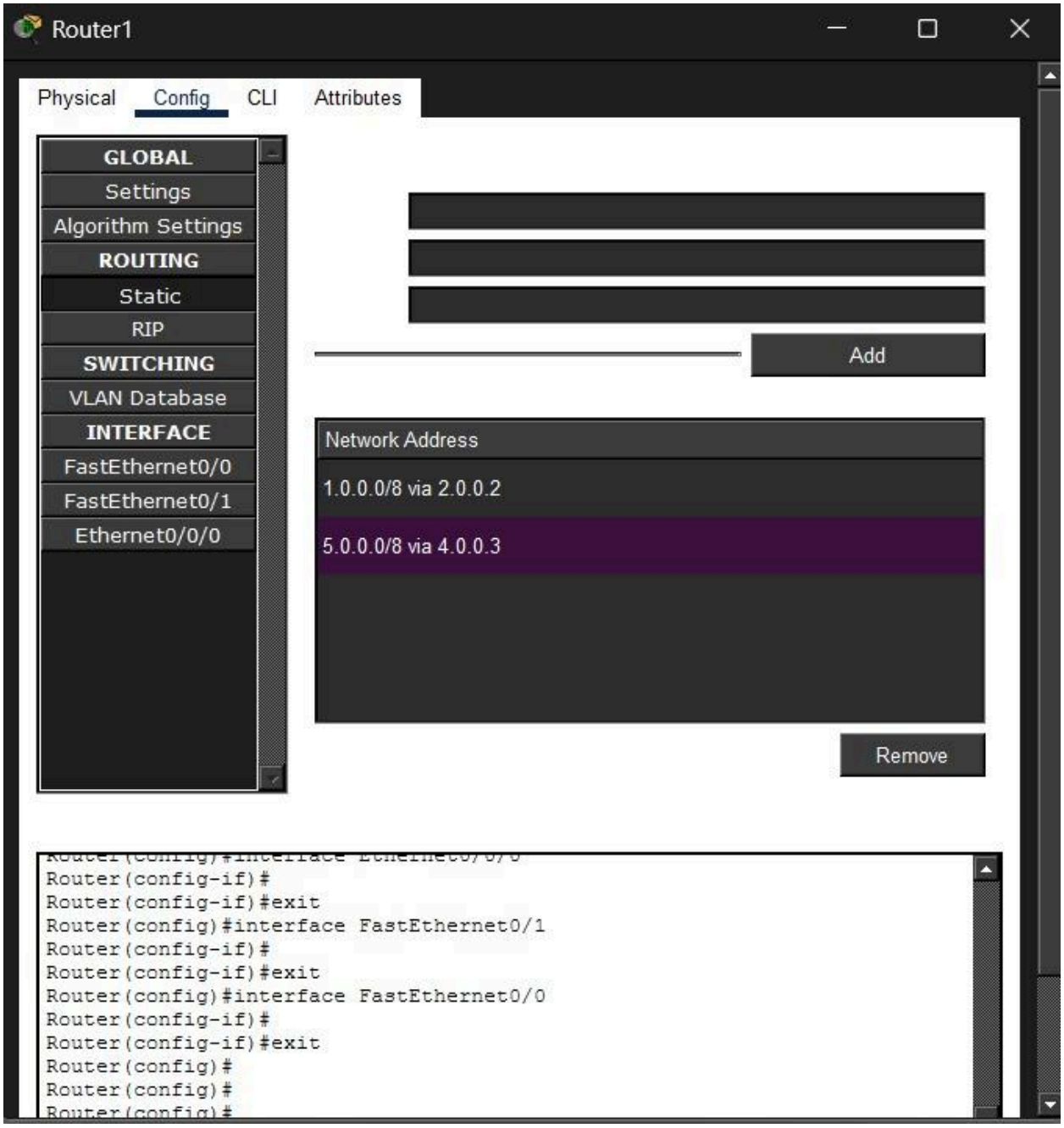

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Add
Network Address
3.0.0.0/8 via 2.0.0.3
4.0.0.0/8 via 2.0.0.3
5.0.0.0/8 via 2.0.0.3
Remove
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#

ROUTER1

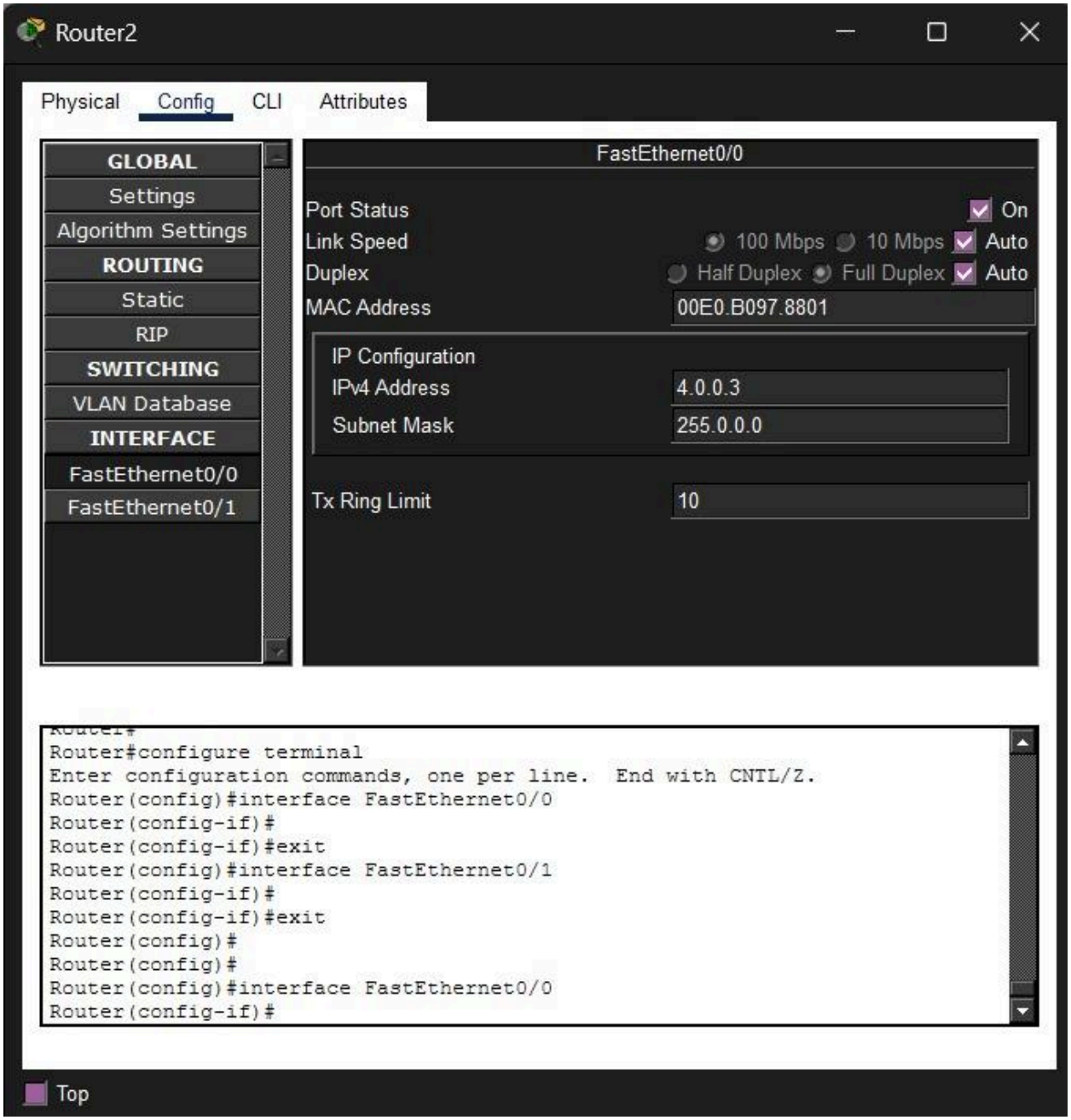

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Ethernet0/0/0
FastEthernet0/0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0060.70E6.8001
IP Configuration
IPv4 Address
2.0.0.3
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Top

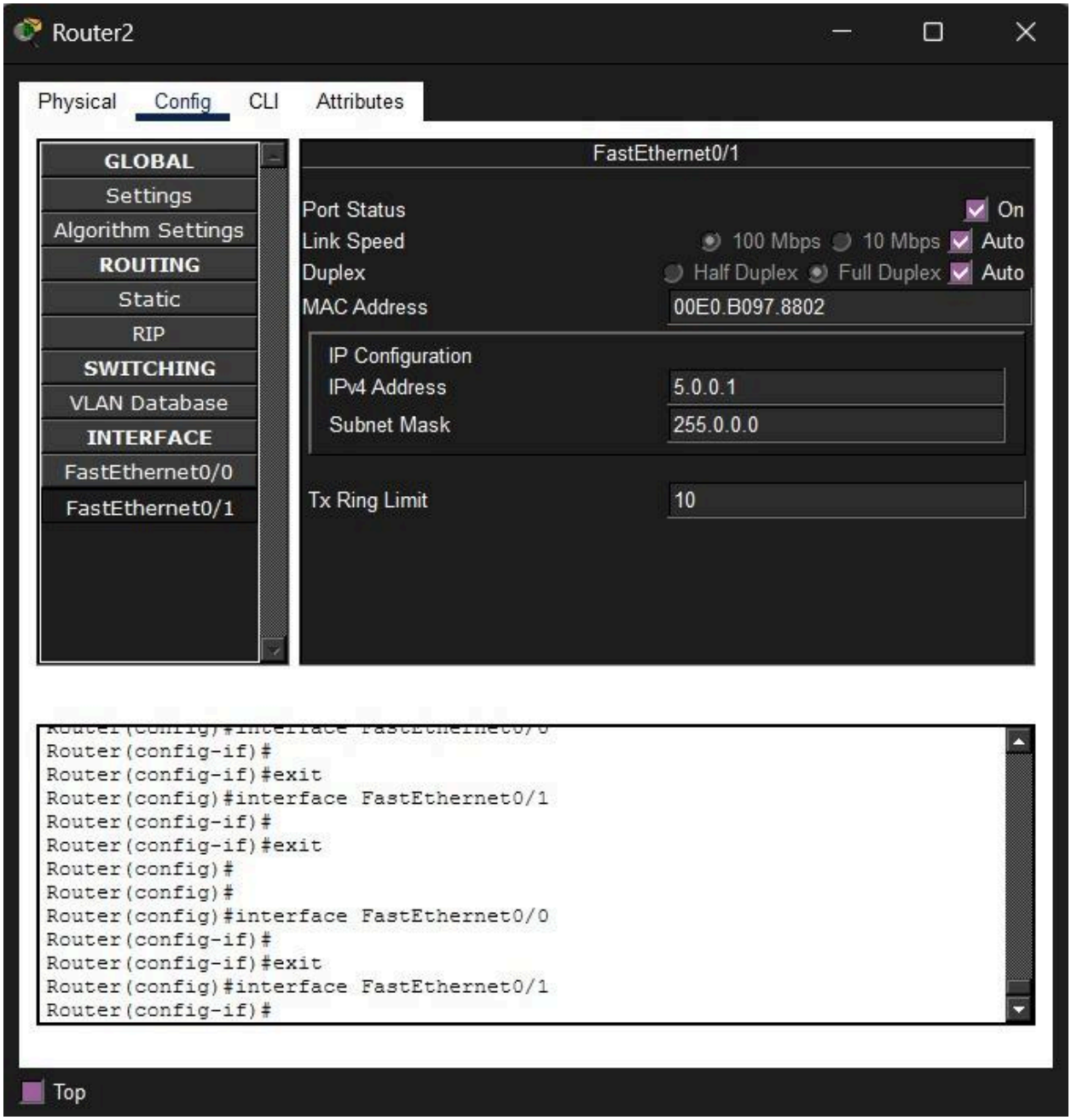

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Ethernet0/0/0
FastEthernet0/1
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0060.70E6.8002
IP Configuration
IPv4 Address
3.0.0.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Top

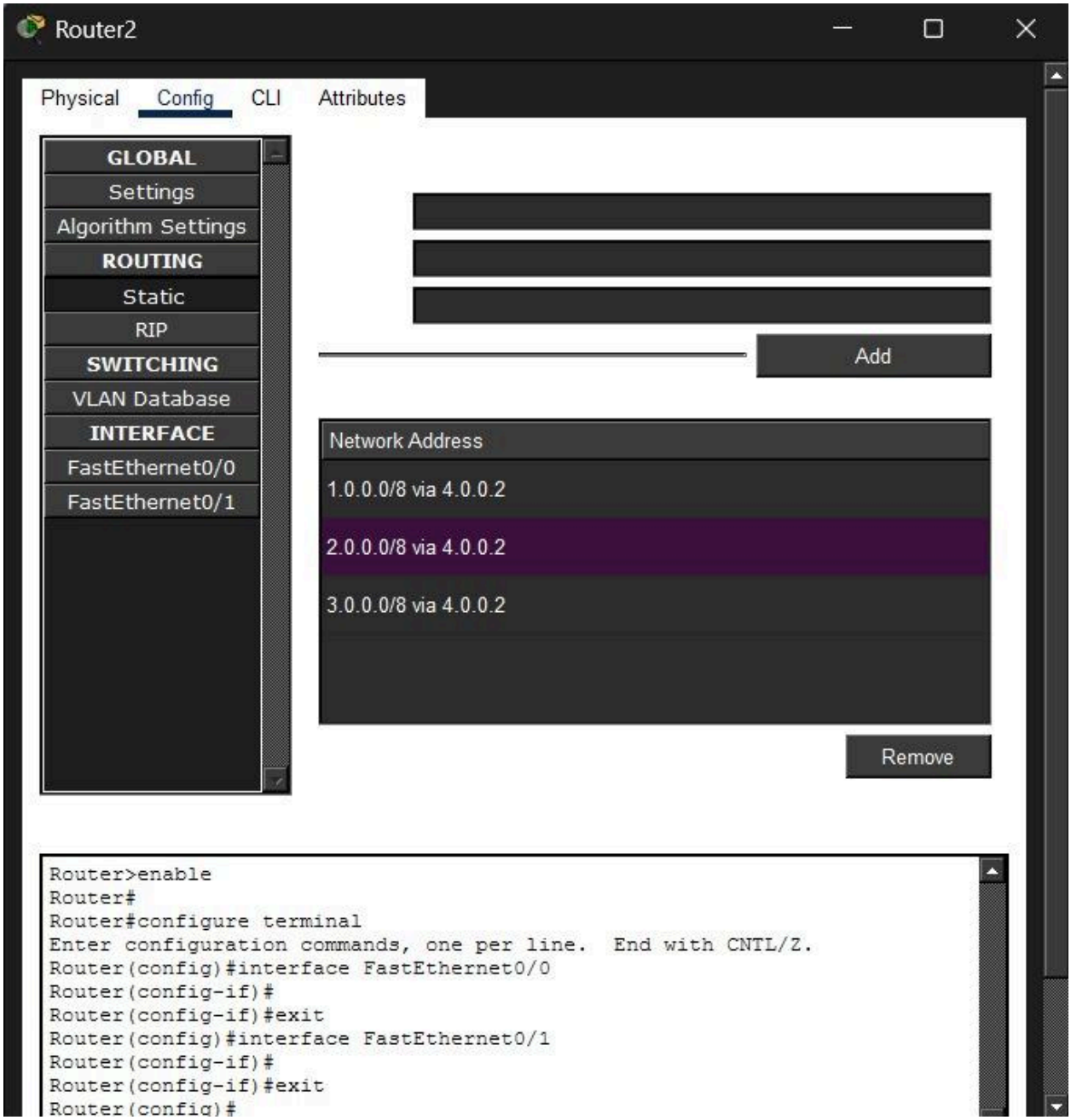

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Ethernet0/0/0
Ethernet0/0/0
Port Status
On
Link Speed
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0010.118C.CE91
IP Configuration
IPv4 Address
4.0.0.2
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Ethernet0/0/0
Router(config-if)#
Top


Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Ethernet0/0/0
Add
Network Address
1.0.0.0/8 via 2.0.0.2
5.0.0.0/8 via 4.0.0.3
Remove
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#

ROUTER2


Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.B097.8801
IP Configuration
IPv4 Address
4.0.0.3
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Top


Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/1
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.B097.8802
IP Configuration
IPv4 Address
5.0.0.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Top



Q3. What is your observation and understanding of this concept that you have implemented?

Answer: I noticed that the routers understand only those networks to which they are directly connected. In order to connect to other networks over long distances, such as from PC0 to PC2, we need to set up Static Routes. This showed me that to connect to other networks, the router requires its own "route" to send data to, or else there would not be any connectivity.

Q4. Why is it necessary to assign an IP address to the newly added interface?

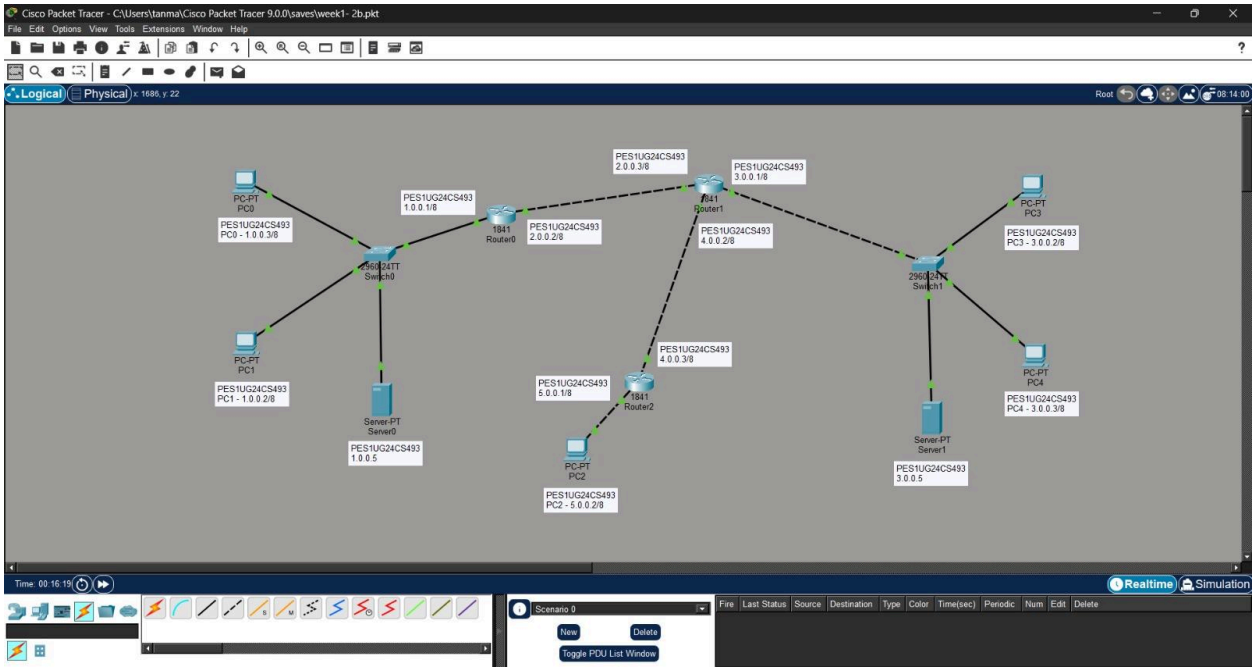
Answer: An interface has to have its own IP address to perform the role of a Default Gateway. This is because, without the IP, the interface will not be able to identify itself in the network, and destinations for the data packets will not exist for the computers (routers, for example).

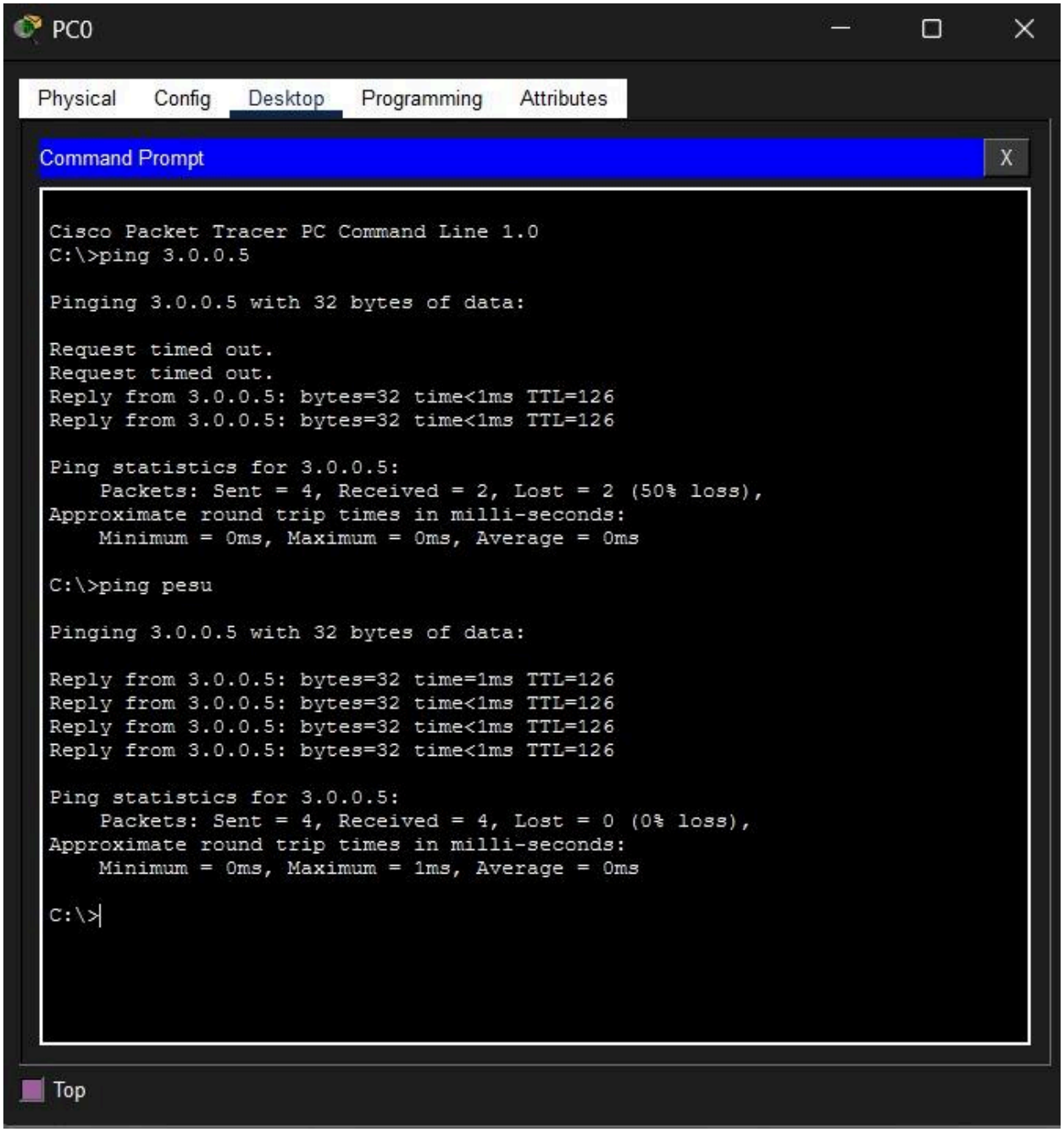
Q5. What is a routing table and why is it important?

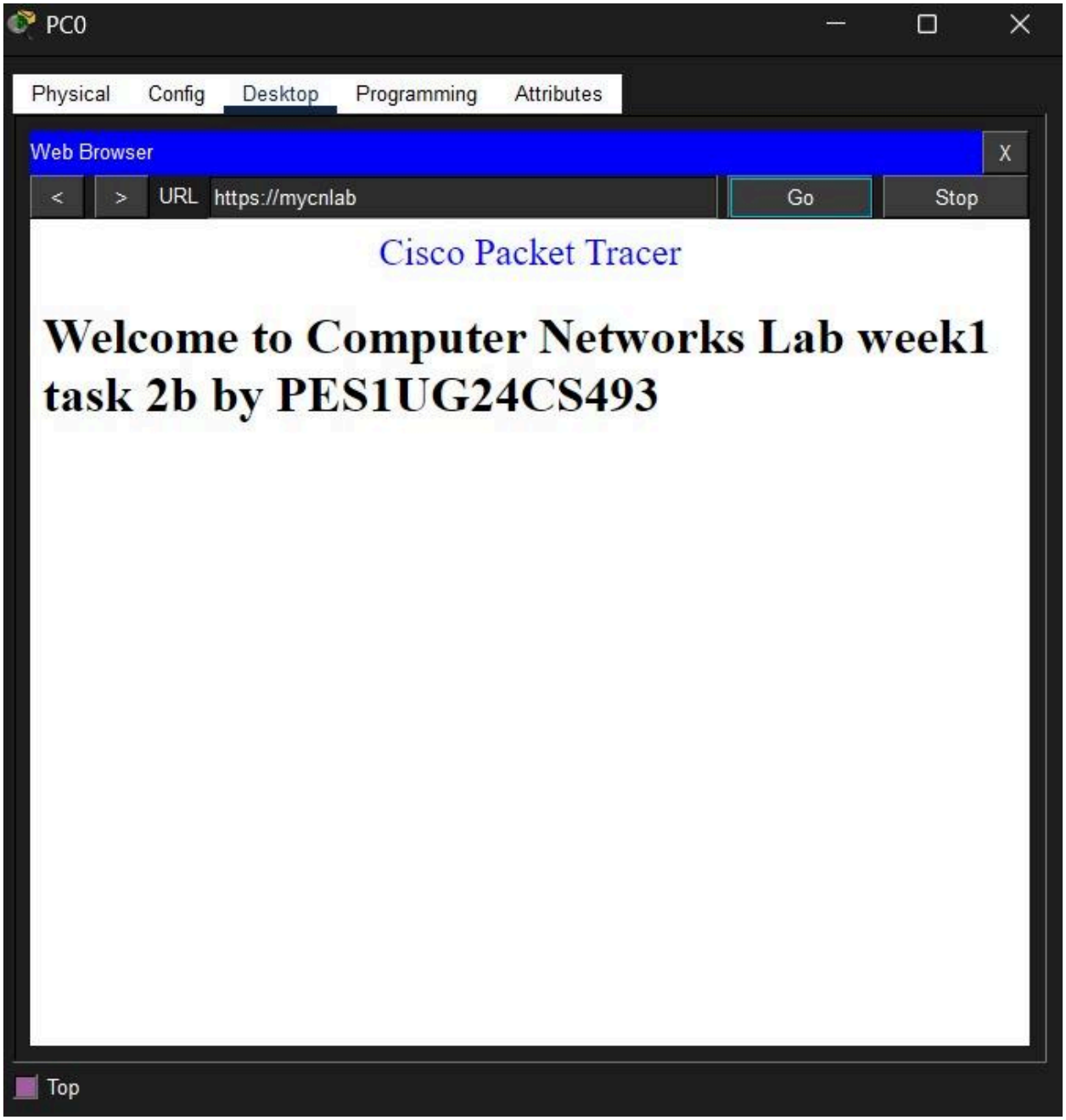
Answer: A Routing Table is like a navigator or a map, which is located inside a router. This contains details of all destination networks and how to send data to these networks. This is because, without it, the router would not know where to send data, and it would discard all packets.

TASK 2b

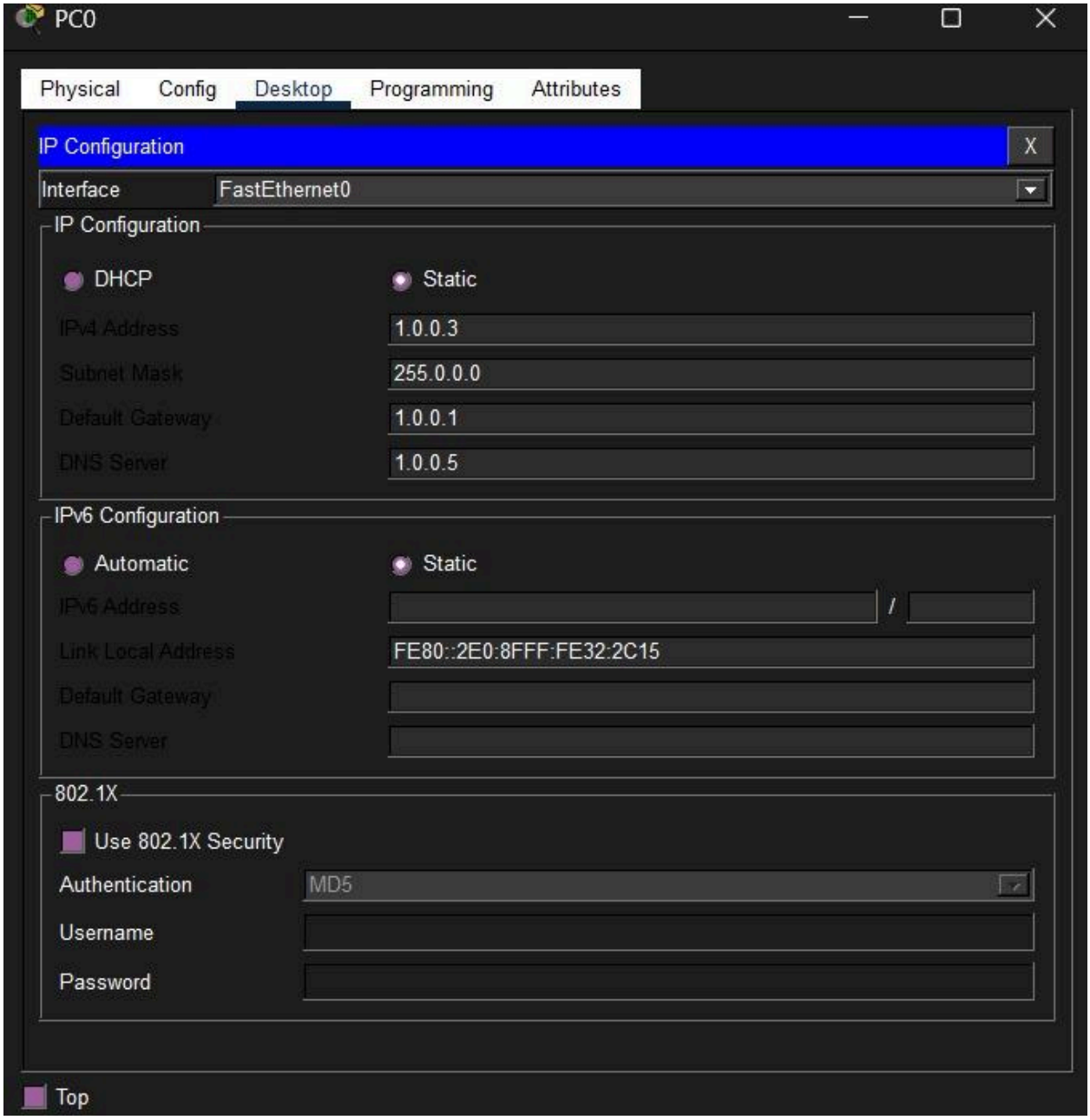
Output



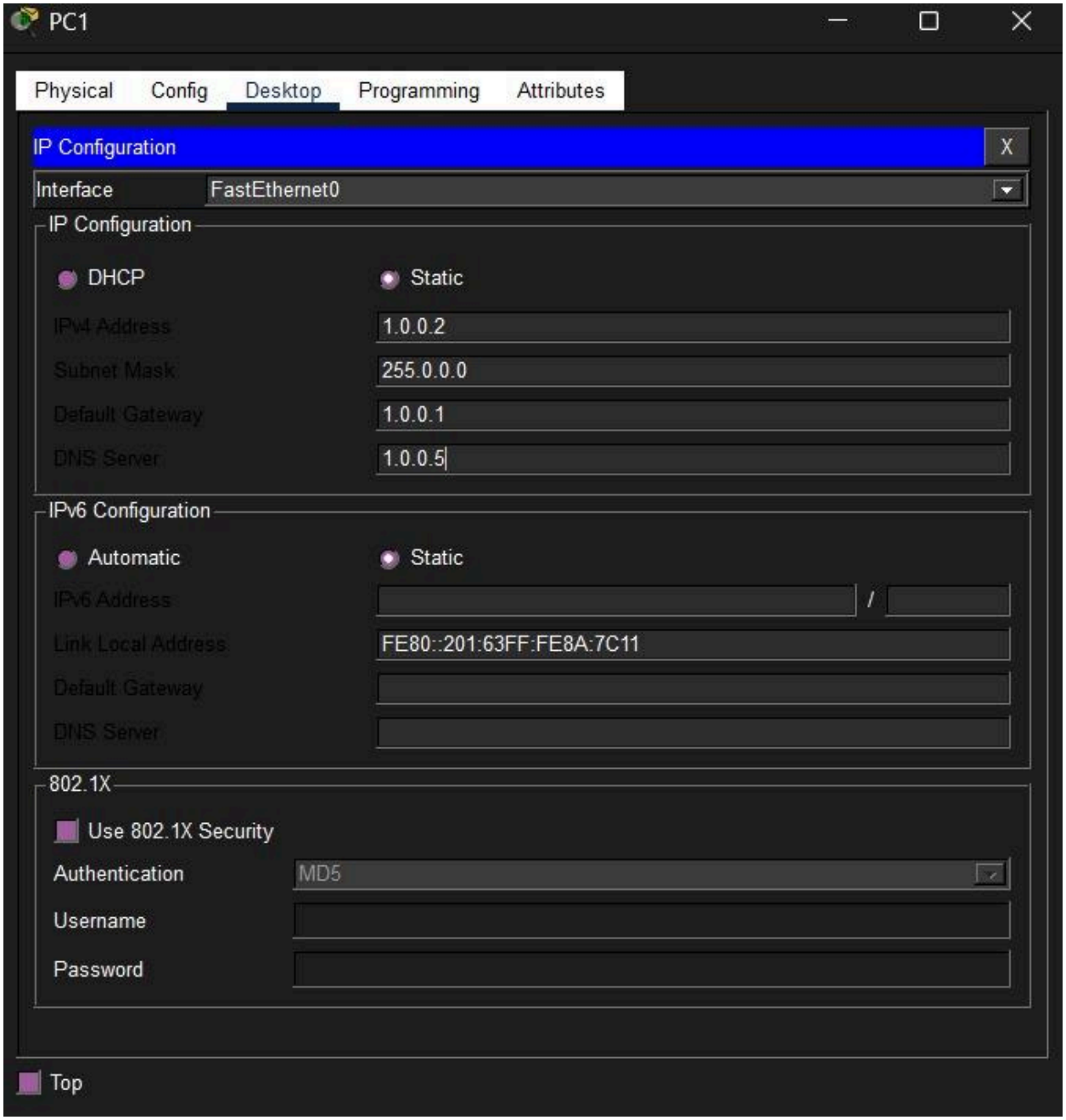

PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3.0.0.5
Pinging 3.0.0.5 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 3.0.0.5: bytes=32 time<1ms TTL=126
Reply from 3.0.0.5: bytes=32 time<1ms TTL=126
Ping statistics for 3.0.0.5:
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping pesu
Pinging 3.0.0.5 with 32 bytes of data:
Reply from 3.0.0.5: bytes=32 time=1ms TTL=126
Reply from 3.0.0.5: bytes=32 time<1ms TTL=126
Reply from 3.0.0.5: bytes=32 time<1ms TTL=126
Reply from 3.0.0.5: bytes=32 time<1ms TTL=126
Ping statistics for 3.0.0.5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top


PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
https://mycnlab
Go
Stop
Cisco Packet Tracer
Welcome to Computer Networks Lab week1 task 2b by PES1UG24CS493
Top

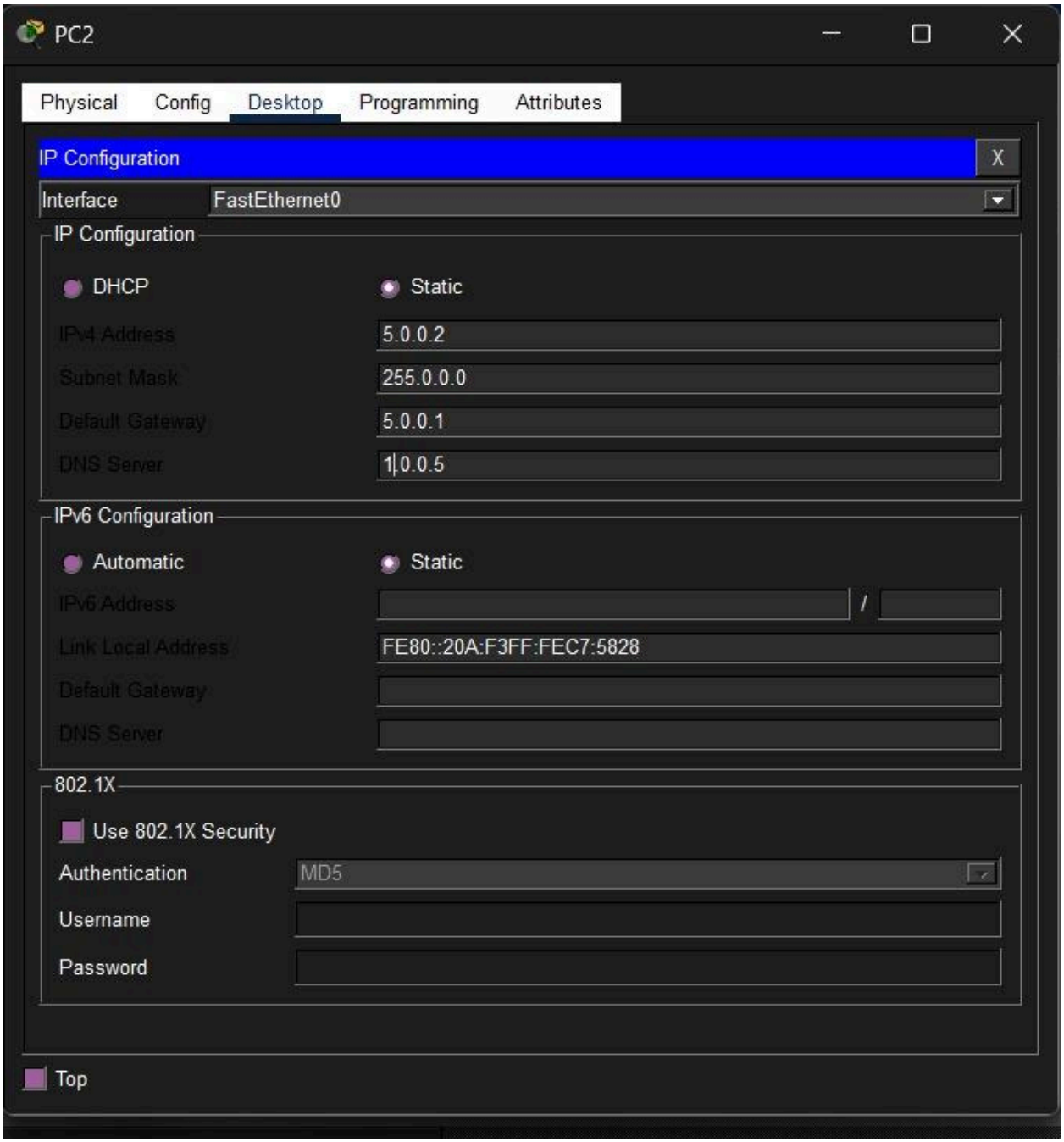
PC0


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
1.0.0.3
Subnet Mask
255.0.0.0
Default Gateway
1.0.0.1
DNS Server
1.0.0.5
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2E0:8FFF:FE32:2C15
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

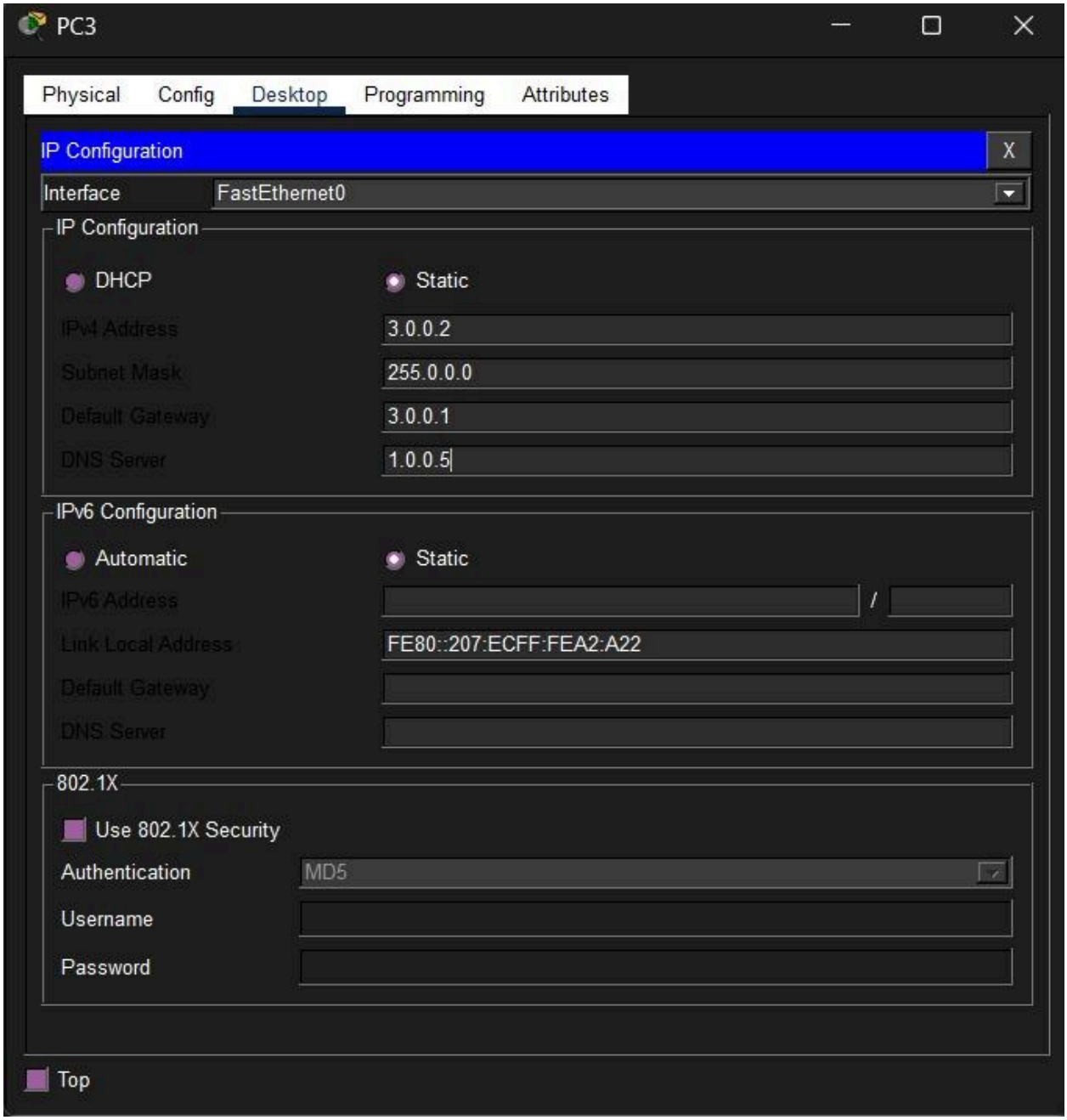
PC1


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
1.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
1.0.0.1
DNS Server
1.0.0.5
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:63FF:FE8A:7C11
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

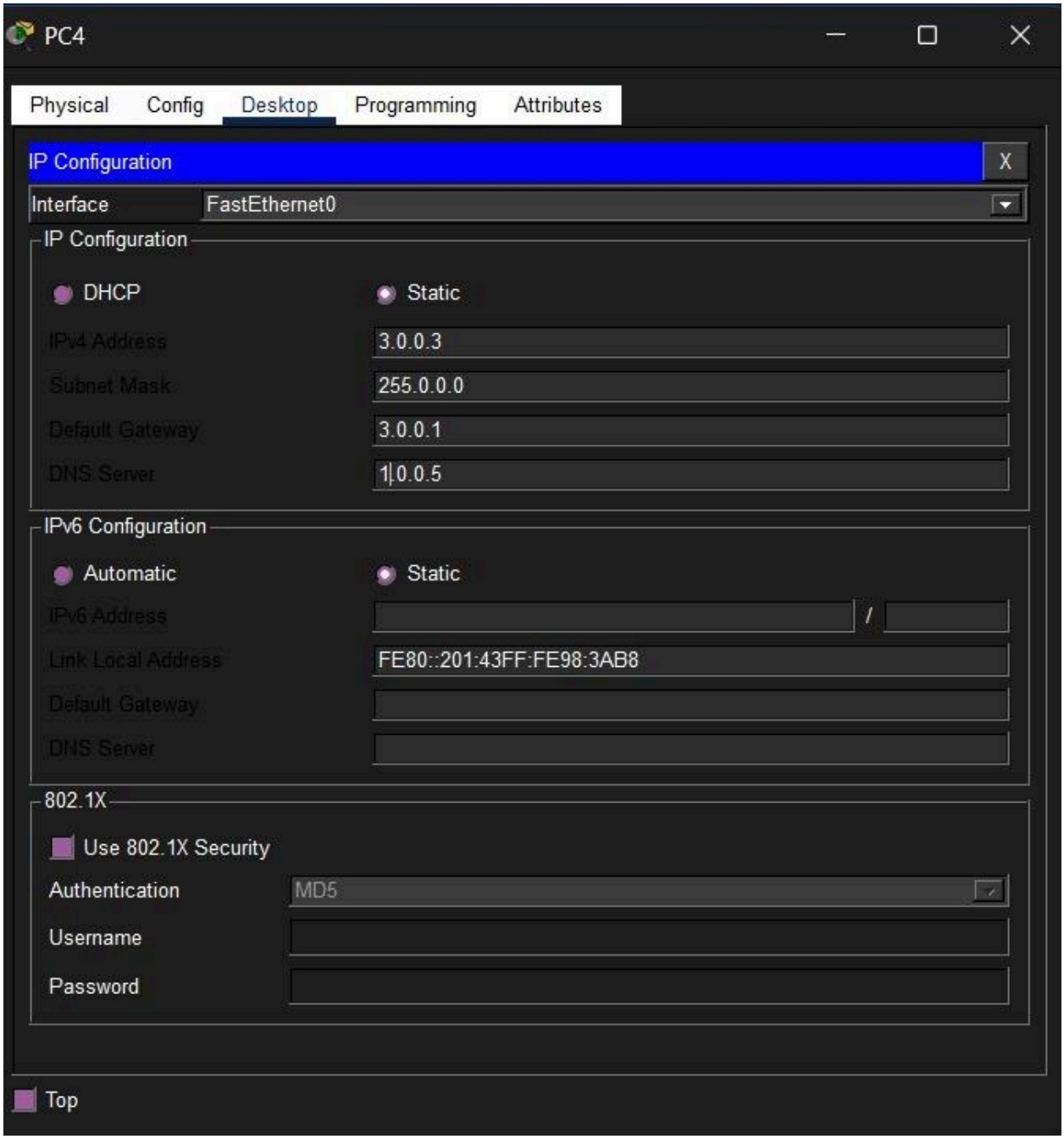
PC2


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
5.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
5.0.0.1
DNS Server
1|0.0.5
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::20A:F3FF:FEC7:5828
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

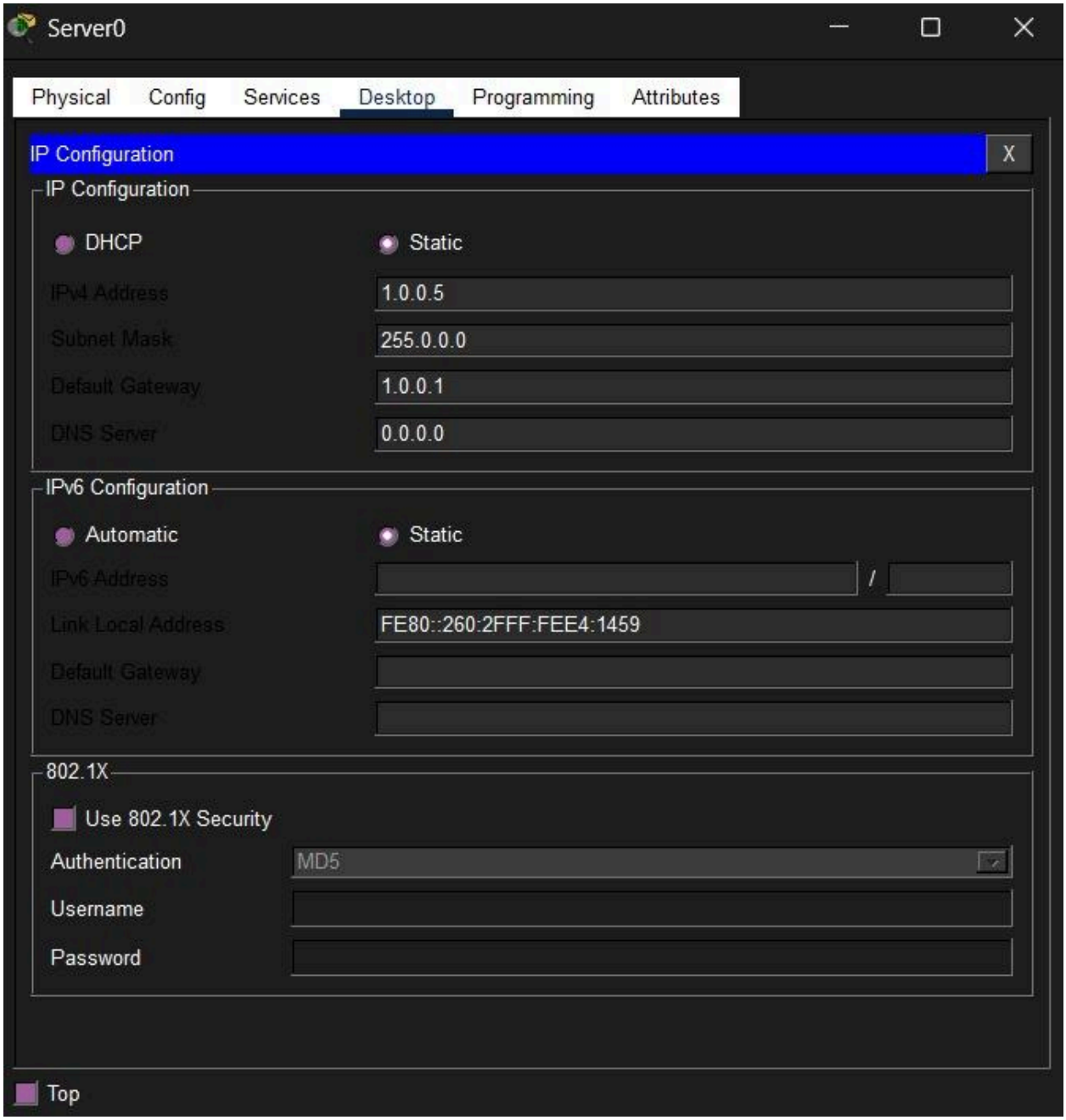
PC3

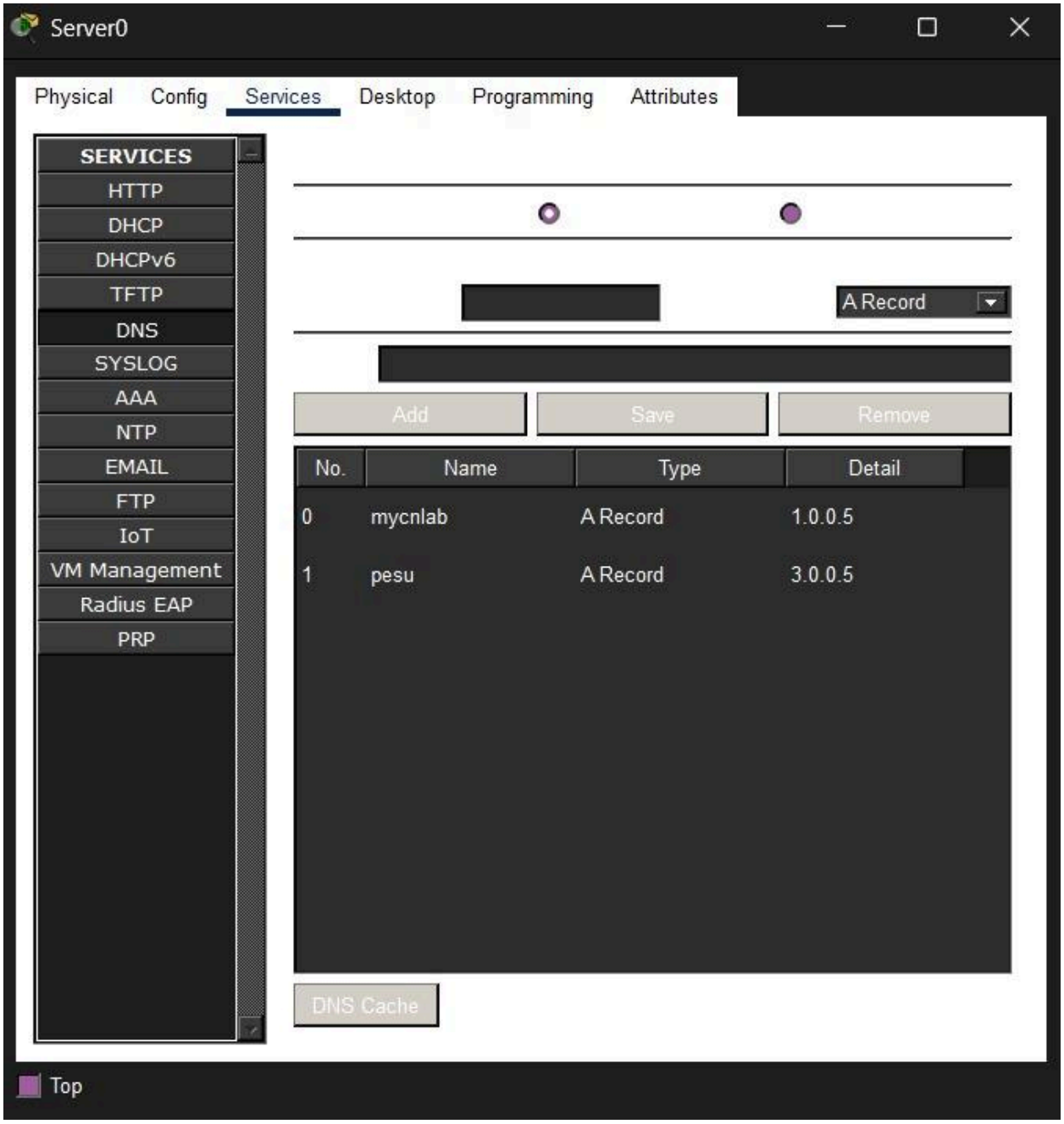

PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
3.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
3.0.0.1
DNS Server
1.0.0.5
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::207:ECFF:FEA2:A22
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

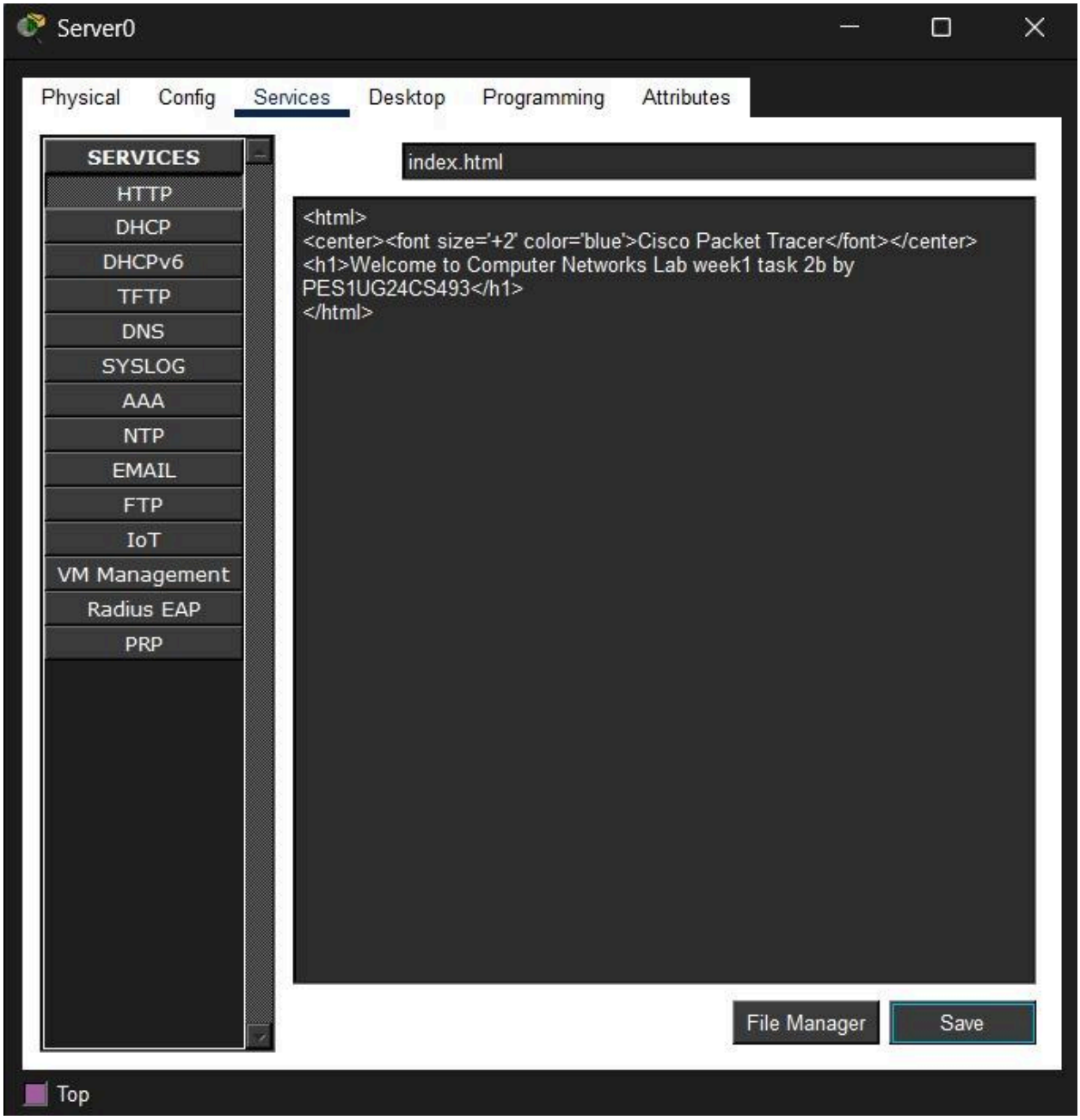
PC4


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
3.0.0.3
Subnet Mask
255.0.0.0
Default Gateway
3.0.0.1
DNS Server
1|0.0.5
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:43FF:FE98:3AB8
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

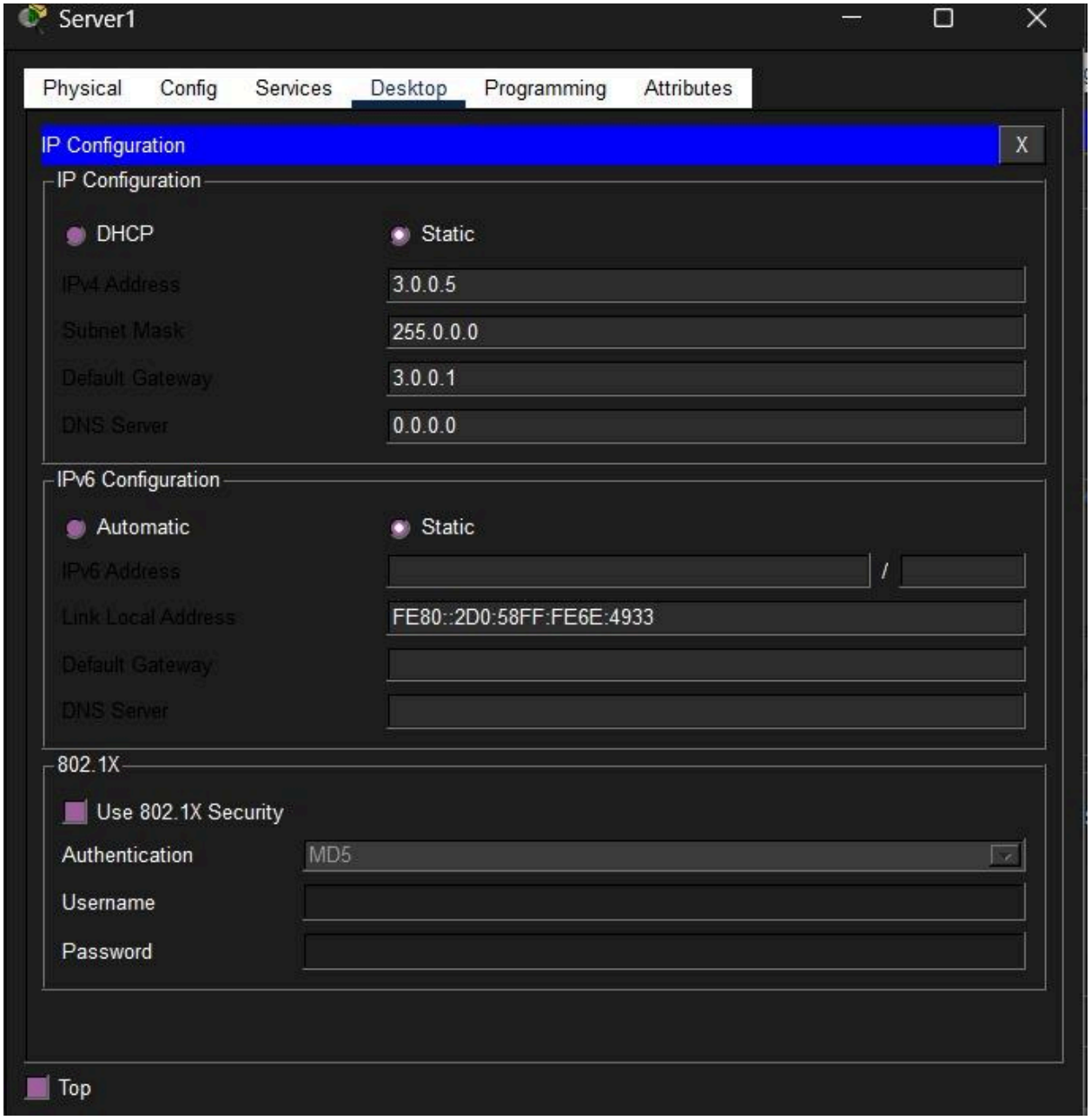
SERVER0


Server0
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
1.0.0.5
Subnet Mask
255.0.0.0
Default Gateway
1.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::260:2FFF:FEE4:1459
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
A Record
Add
Save
Remove
No.
Name
Type
Detail
0
mycnlab
A Record
1.0.0.5
1
pesu
A Record
3.0.0.5
DNS Cache
Top


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
index.html
<html>
Cisco Packet Tracer
<h1>Welcome to Computer Networks Lab week1 task 2b by
PES1UG24CS493</h1>
</html>
File Manager
Save
Top

SERVER1



2. What is your observation and understanding of this concept that you have implemented?

Answer: One observation I made was that with a DNS server, it is possible to connect to the web server not by IP address, but by name, for example, `pesu`. This made me aware: DNS automatically translates convenient names to IP addresses that computers must connect to.

3. Why must each PC be configured with the DNS server's IP address?

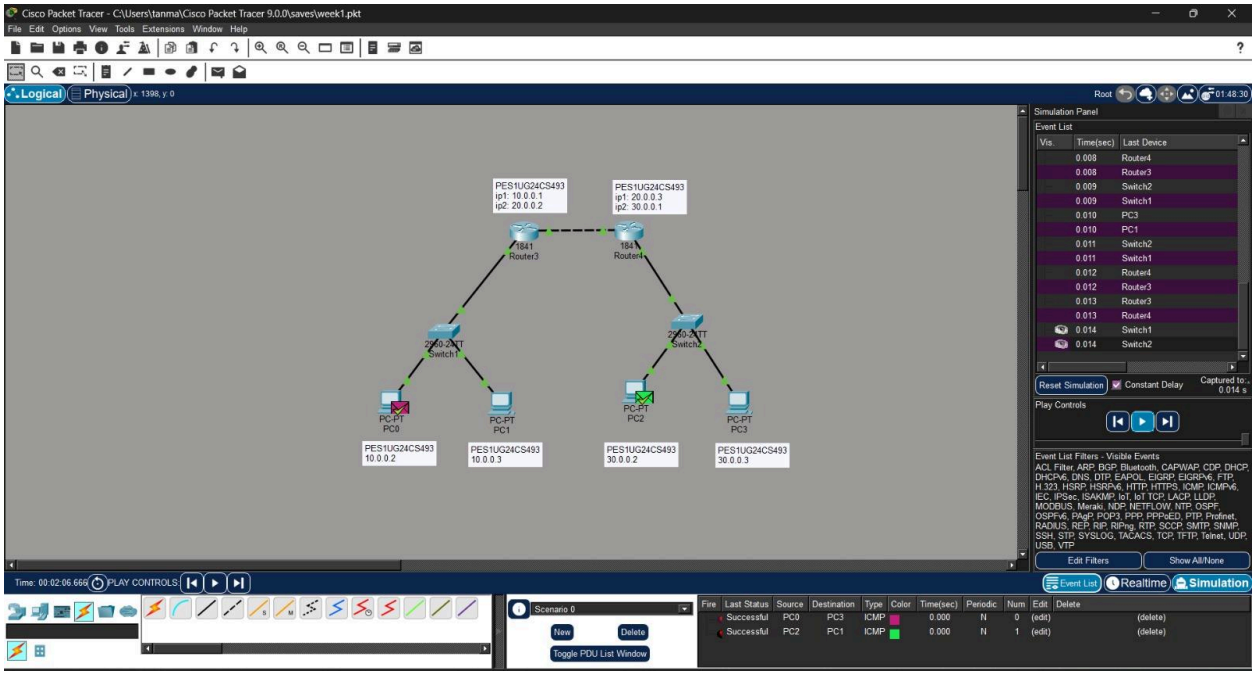
Answer: The PC must have information about **whom to query for the translation** of names to IPs. If we do not store the address of the DNS Server in the settings of the PC, the computer itself would not know where to send the query, and the websites would not be loaded by names.

4. What is the role of a DNS server in a network?

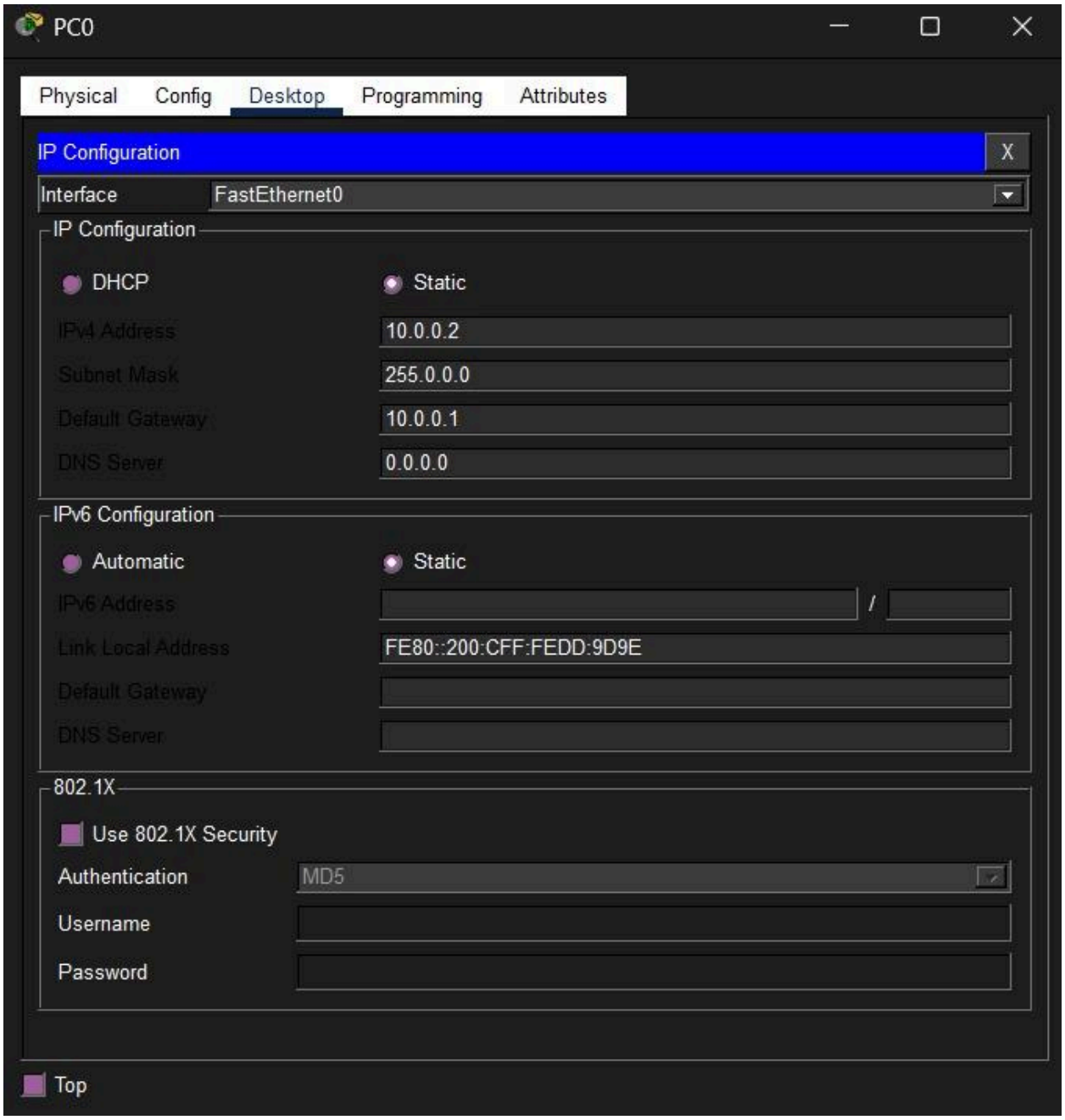
Answer: A DNS server is like an electronic phone book. The primary function of a DNS server is to convert human-readable names (URLs) to machine-readable IP addresses so that computers can find each other on the network.

TASK 1a

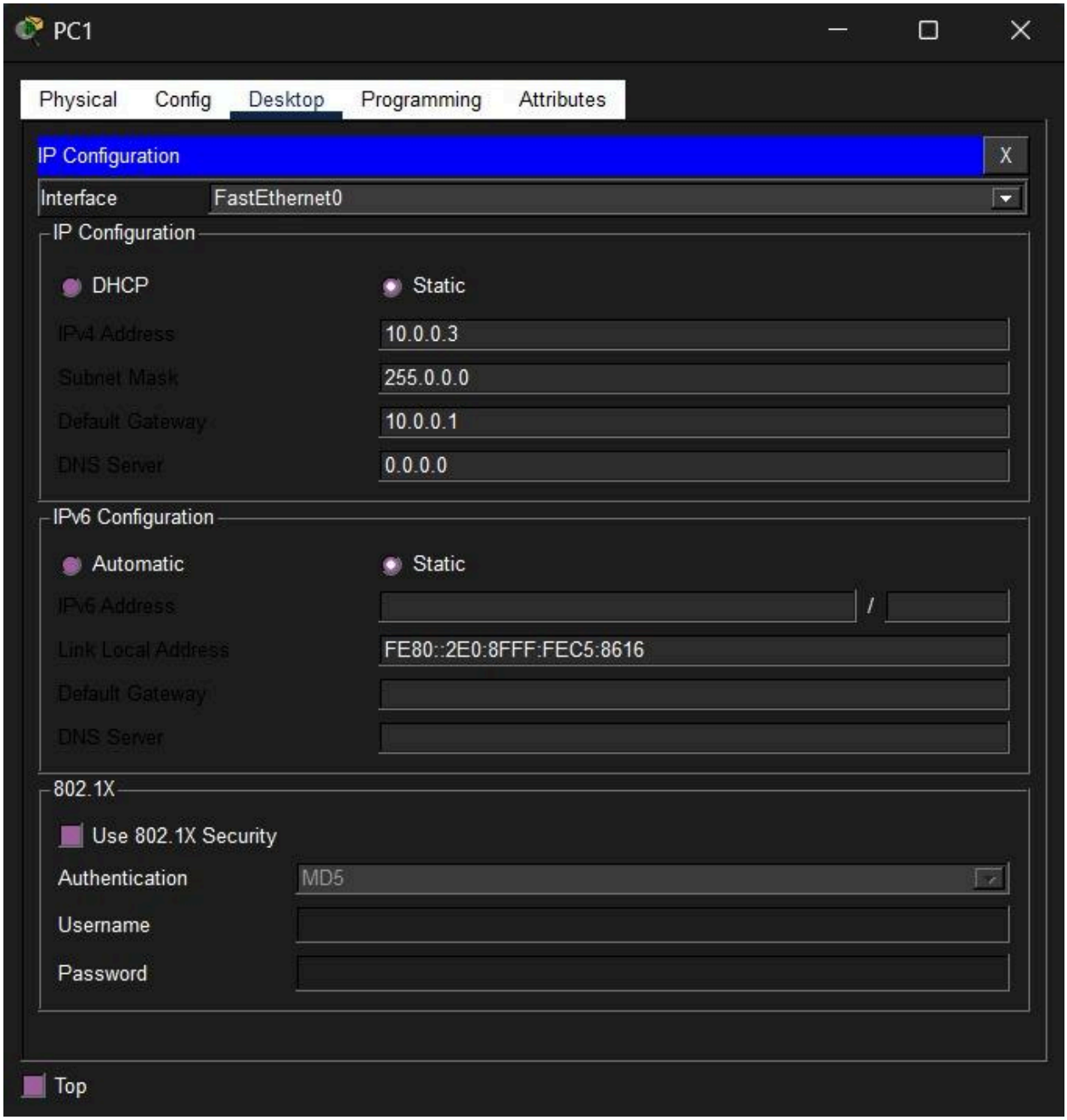
Output



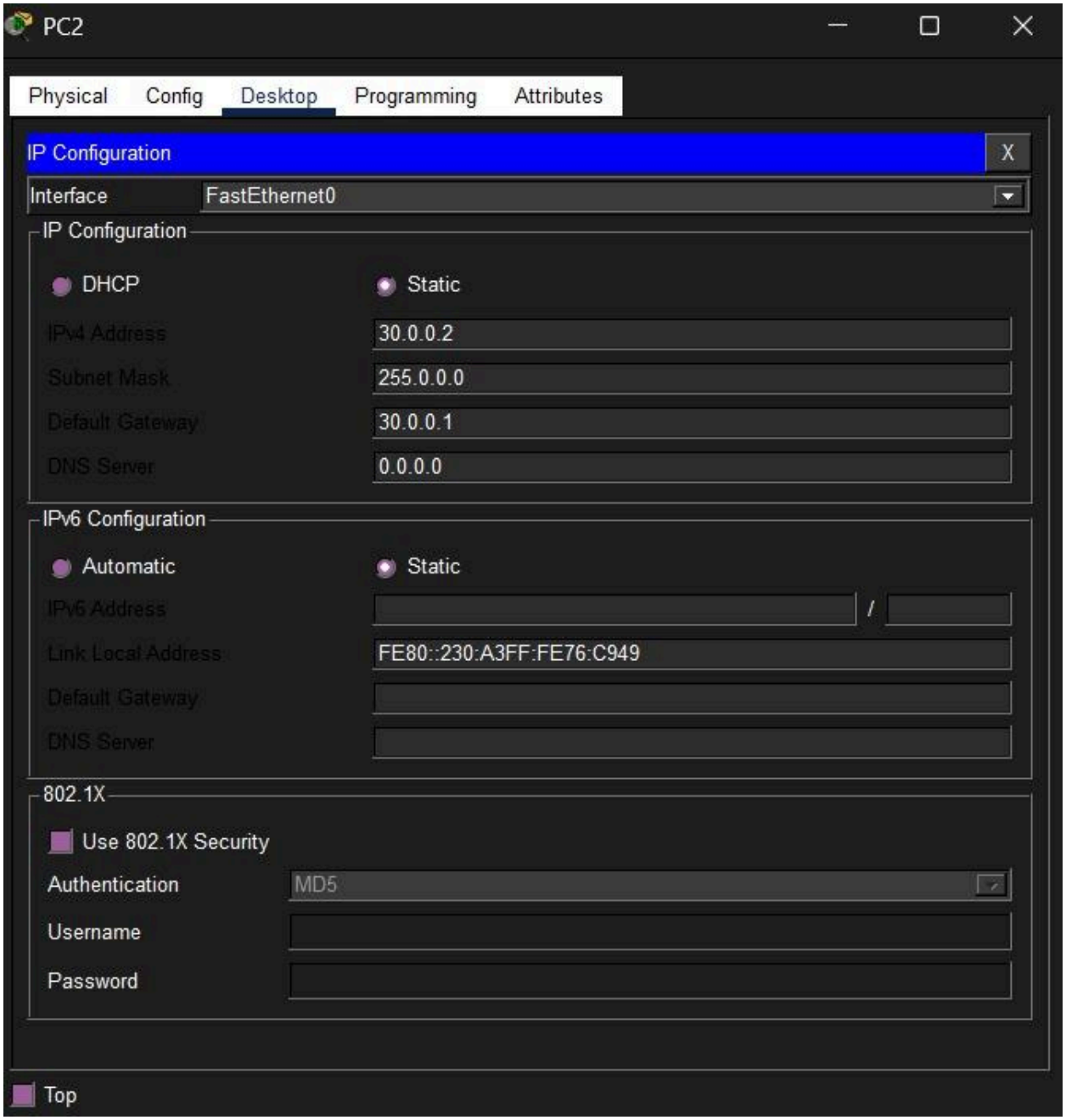
PC0


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
10.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::200:CFF:FEDD:9D9E
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

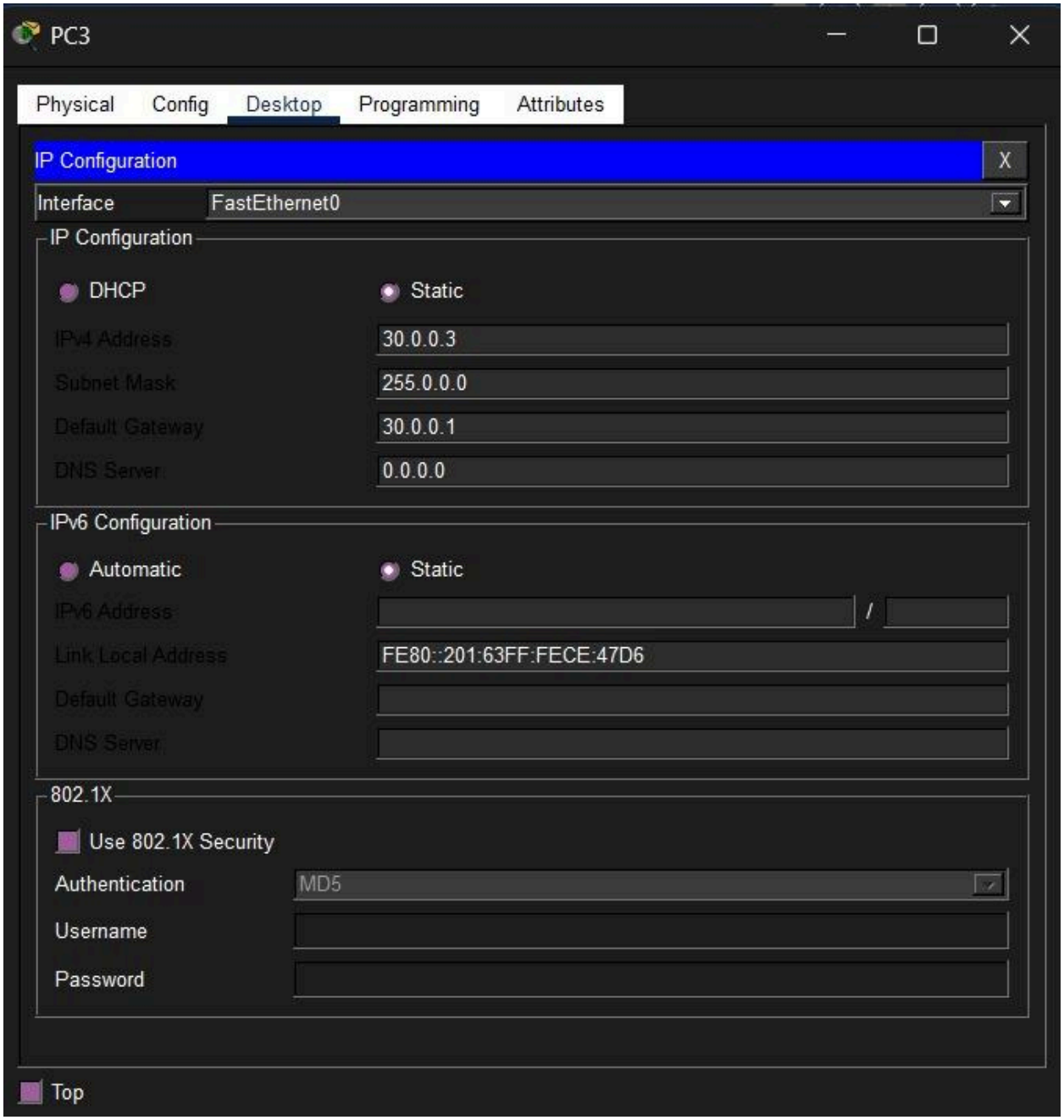
PC1


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.3
Subnet Mask
255.0.0.0
Default Gateway
10.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2E0:8FFF:FEC5:8616
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

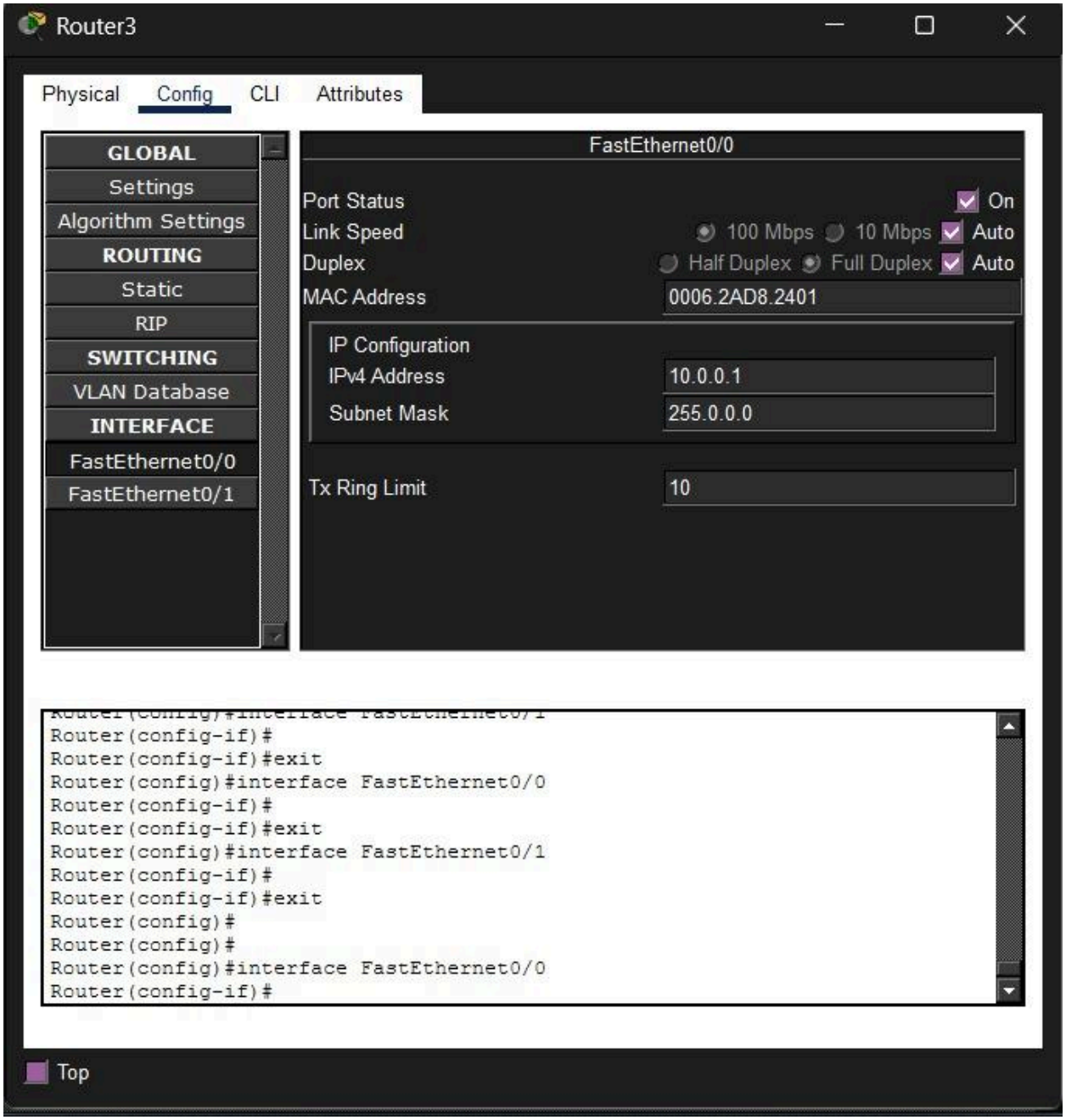
PC2

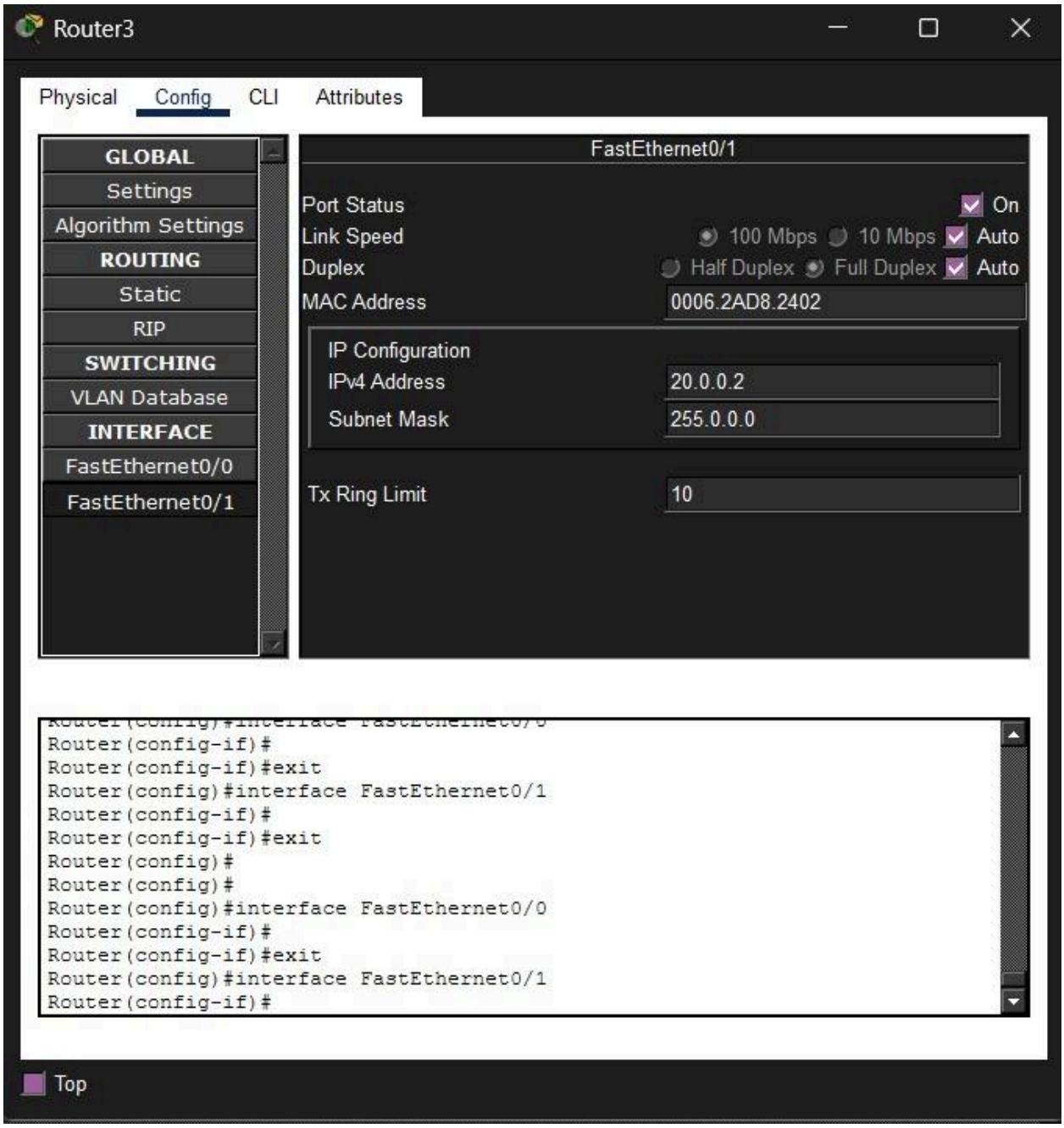

PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
30.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
30.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::230:A3FF:FE76:C949
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

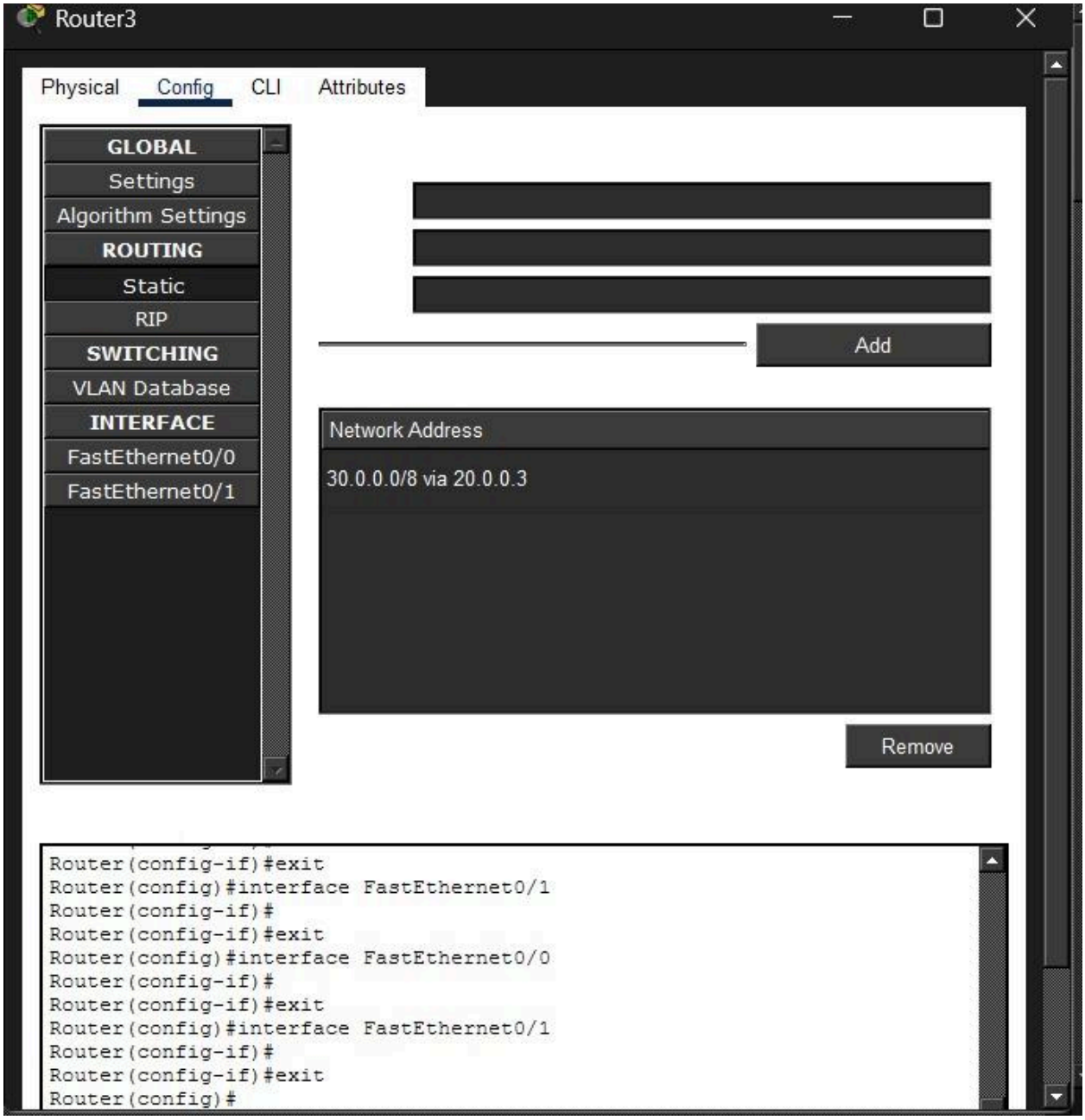
PC3


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
30.0.0.3
Subnet Mask
255.0.0.0
Default Gateway
30.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:63FF:FECE:47D6
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

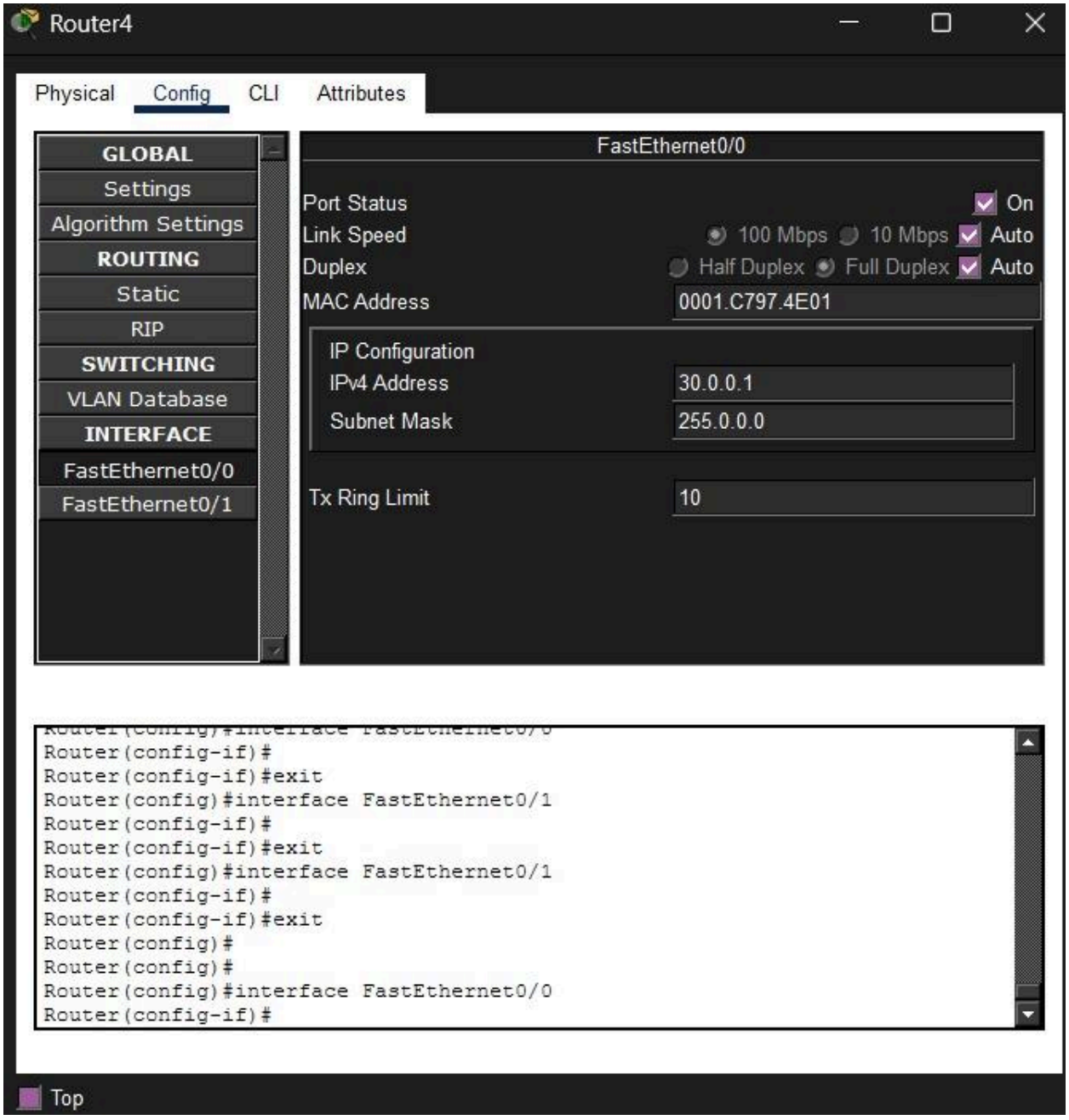
ROUTER3

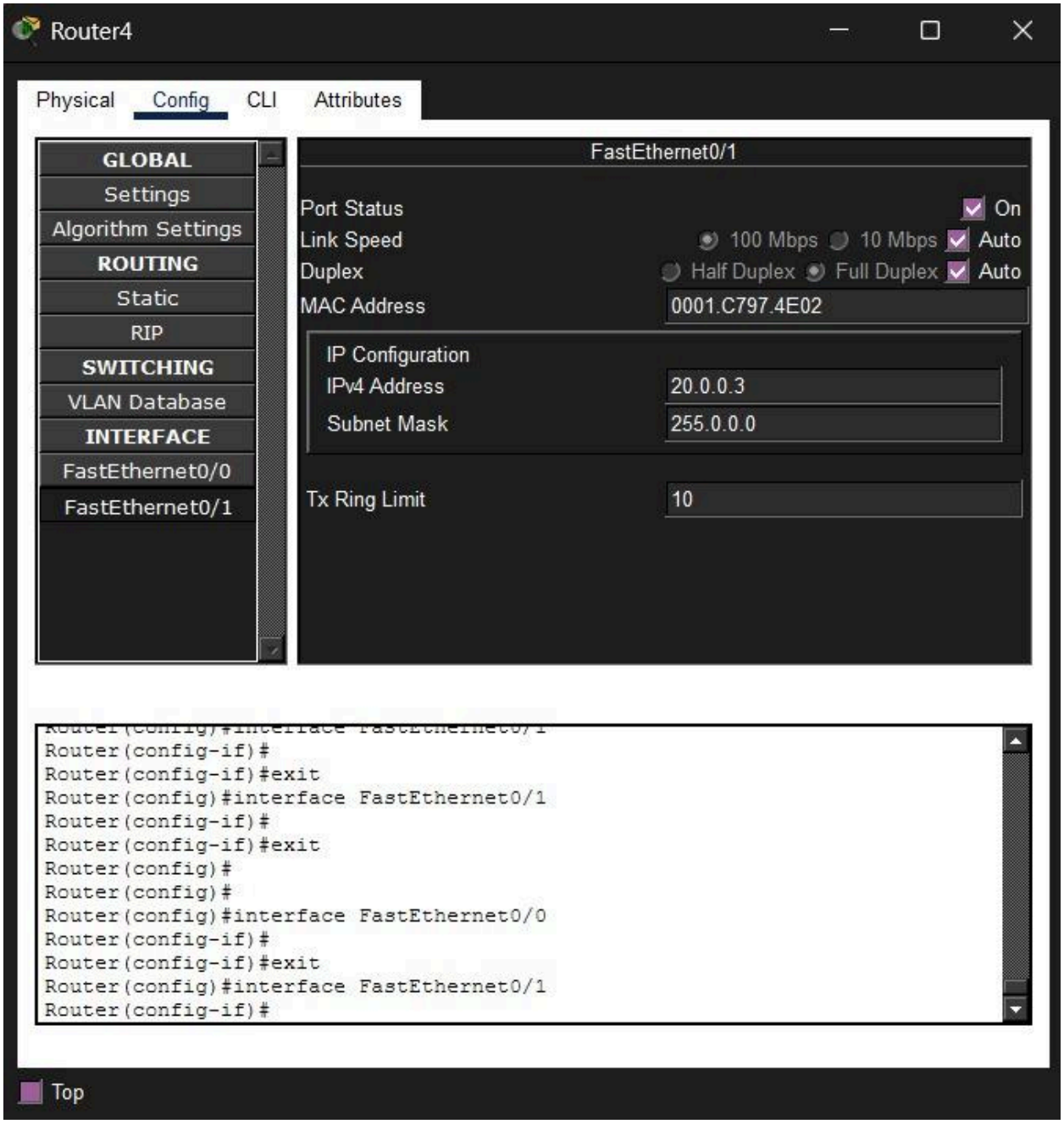

Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0006.2AD8.2401
IP Configuration
IPv4 Address
10.0.0.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Top

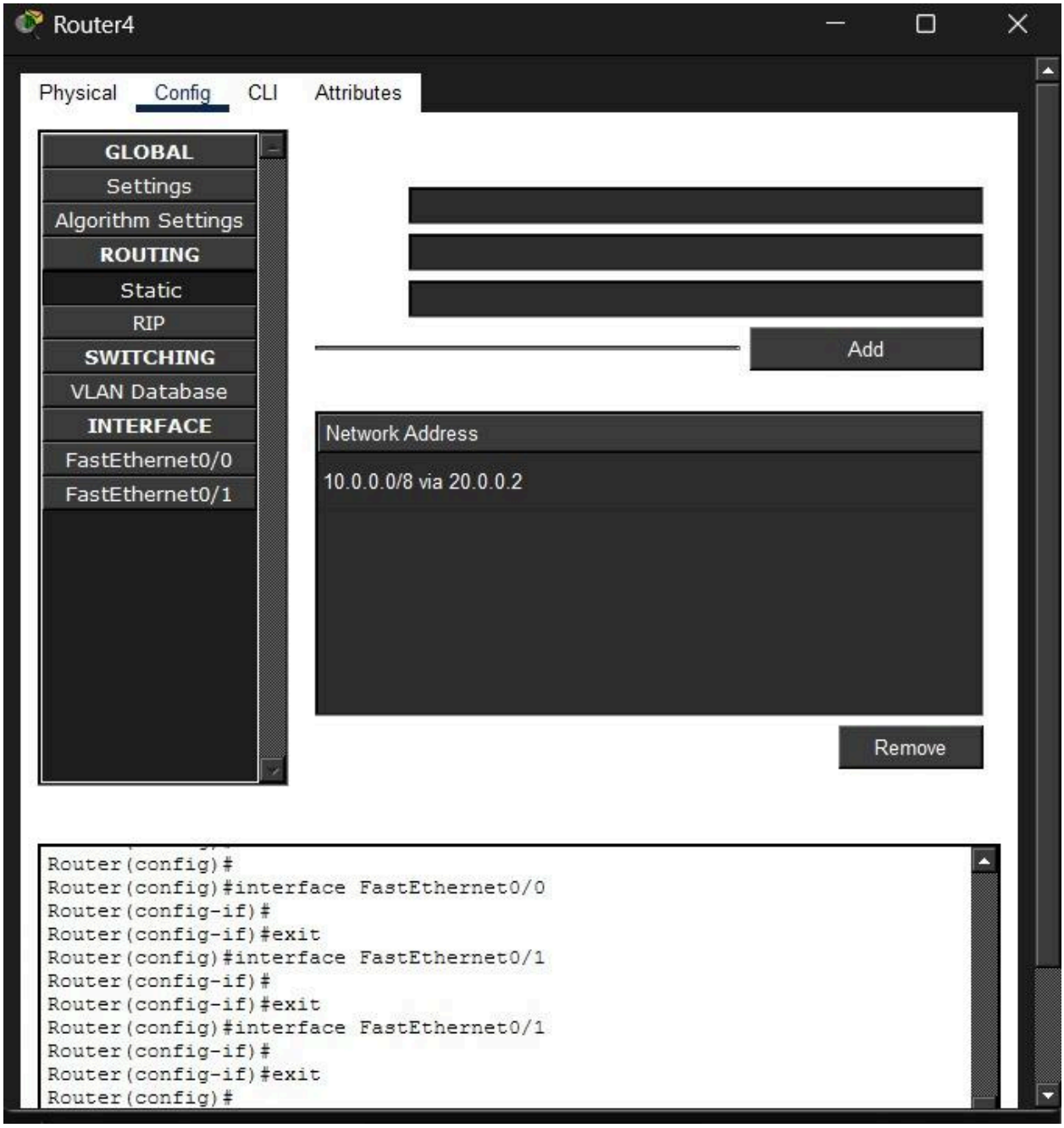

Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/1
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0006.2AD8.2402
IP Configuration
IPv4 Address
20.0.0.2
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Top


Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Add
Network Address
30.0.0.0/8 via 20.0.0.3
Remove
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#

ROUTER4

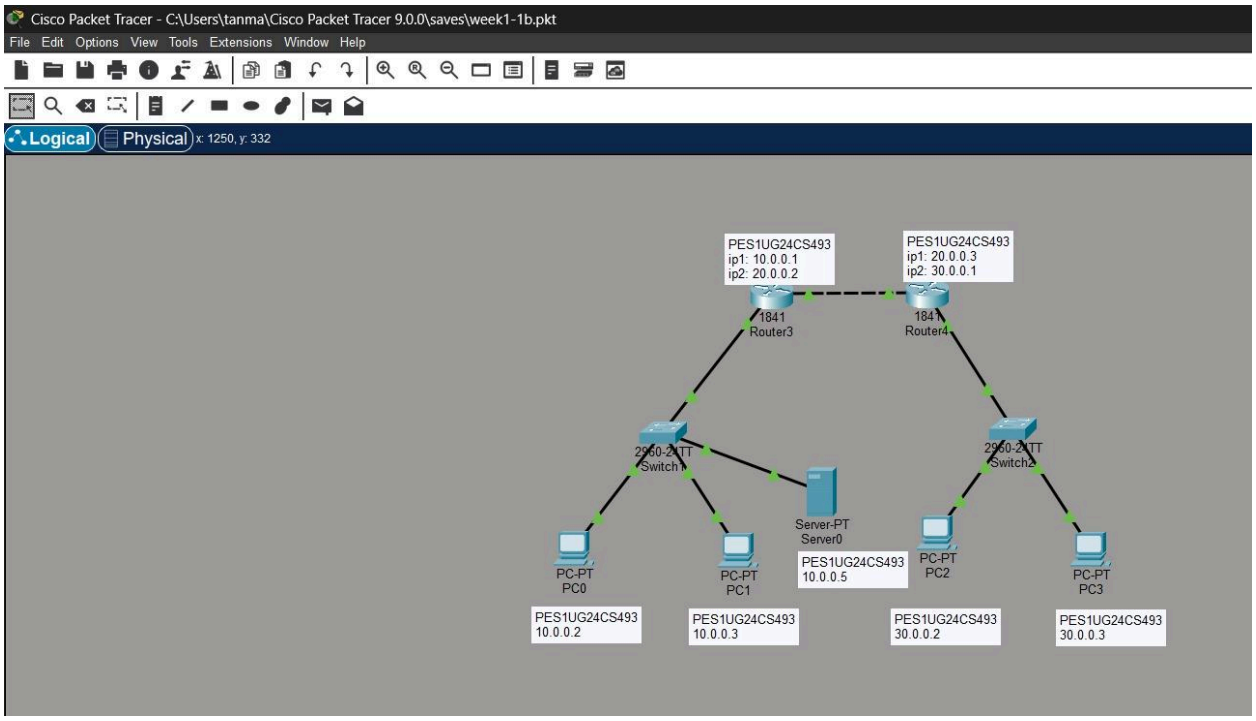

Router4
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.C797.4E01
IP Configuration
IPv4 Address
30.0.0.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Top

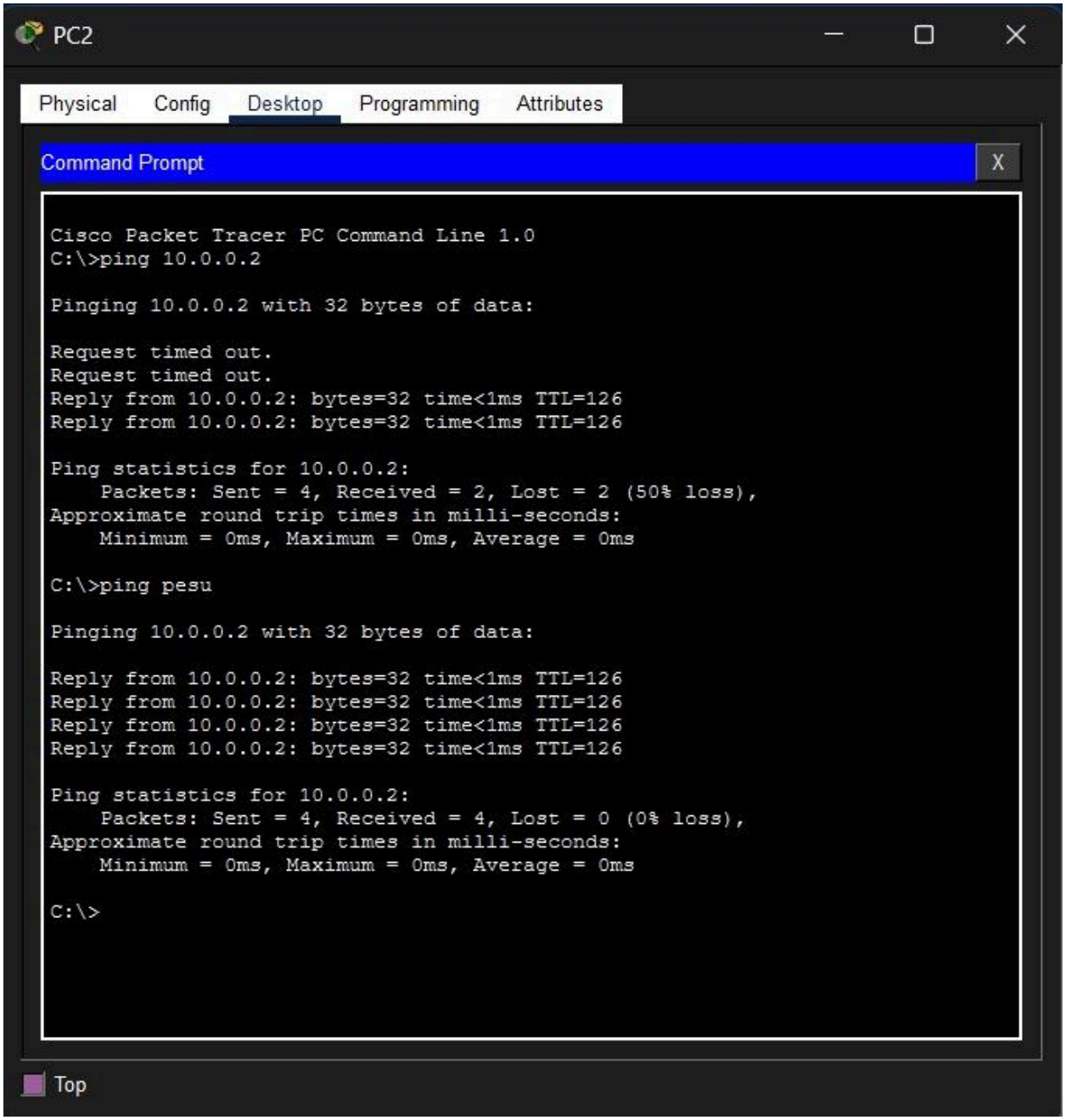

Router4
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/1
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.C797.4E02
IP Configuration
IPv4 Address
20.0.0.3
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Top

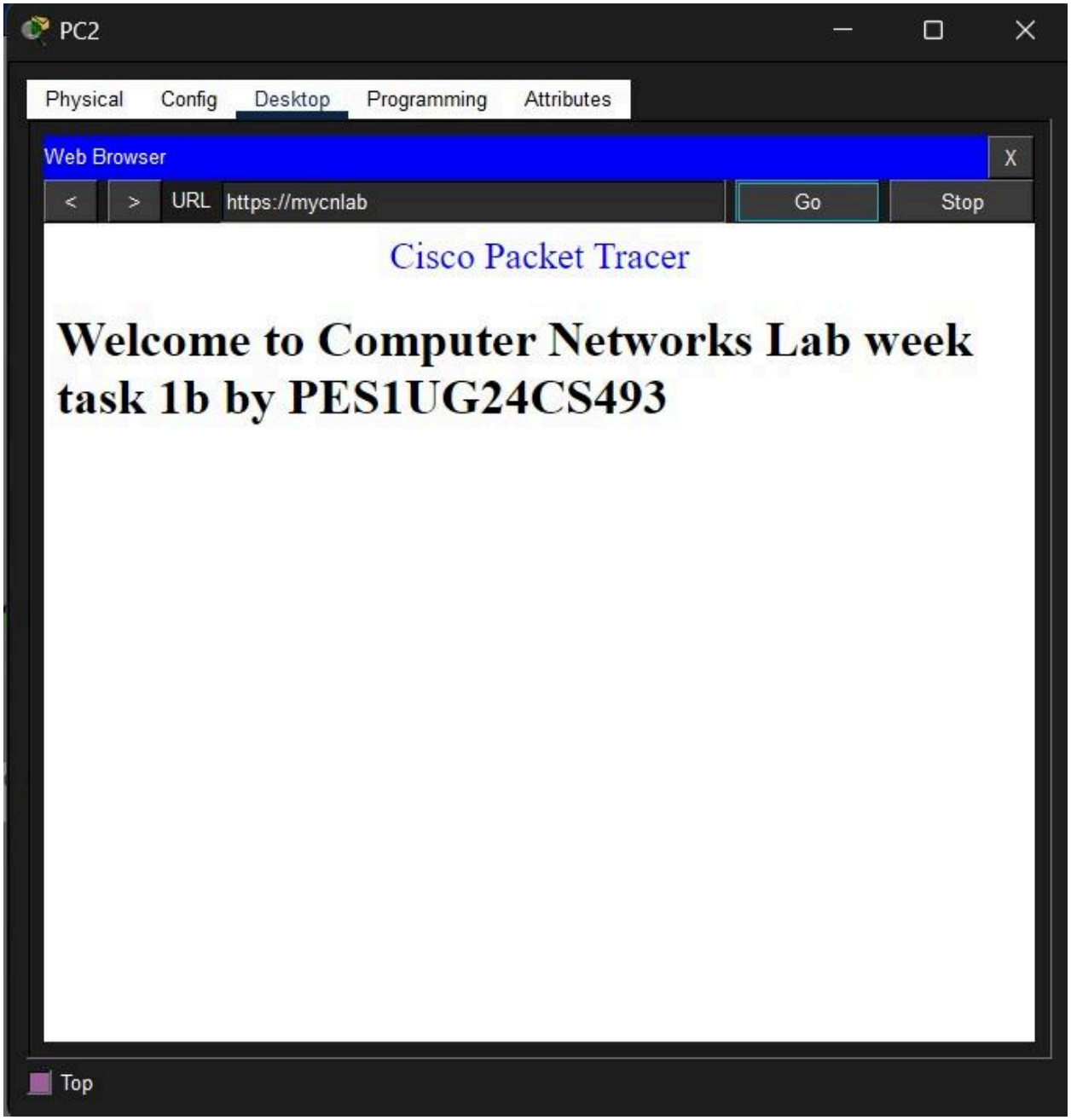

Router4
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Add
Network Address
10.0.0.0/8 via 20.0.0.2
Remove
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#

TASK 1b

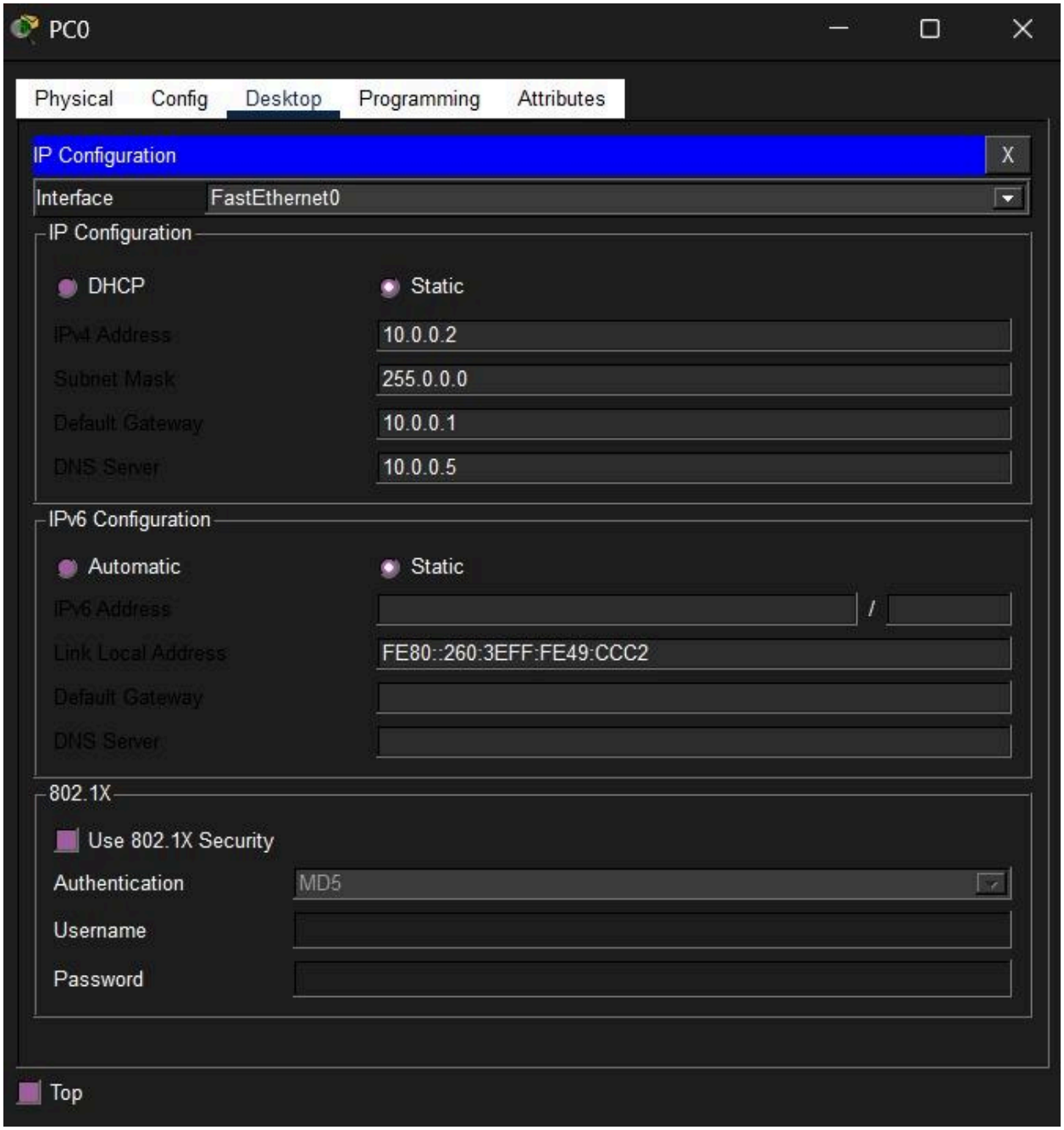
Output


Cisco Packet Tracer - C:\Users\tanma\Cisco Packet Tracer 9.0.0\saves\week1-1b.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1250, y: 332
PES1UG24CS493
ip1: 10.0.0.1
ip2: 20.0.0.2
PES1UG24CS493
ip1: 20.0.0.3
ip2: 30.0.0.1
1841
Router3
1841
Router4
2960-24TT
Switch1
2960-24TT
Switch2
Server-PT
Server0
PES1UG24CS493
10.0.0.5
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PES1UG24CS493
10.0.0.2
PES1UG24CS493
10.0.0.3
PES1UG24CS493
30.0.0.2
PES1UG24CS493
30.0.0.3

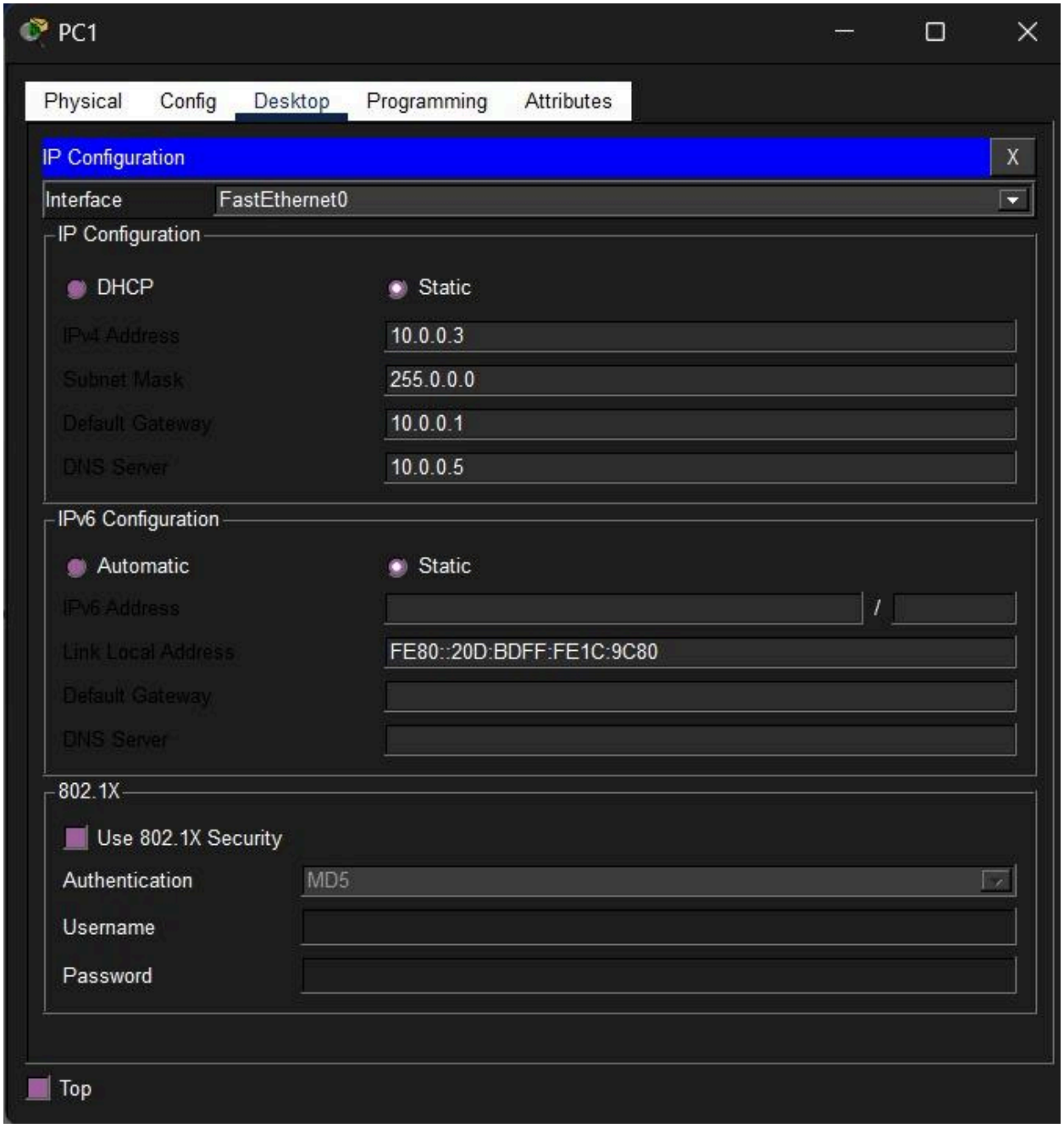

PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.2
Pinging 10.0.0.2 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Ping statistics for 10.0.0.2:
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping pesu
Pinging 10.0.0.2 with 32 bytes of data:
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Ping statistics for 10.0.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top


PC2
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
https://mycnlab
Go
Stop
Cisco Packet Tracer
Welcome to Computer Networks Lab week task 1b by PES1UG24CS493
Top

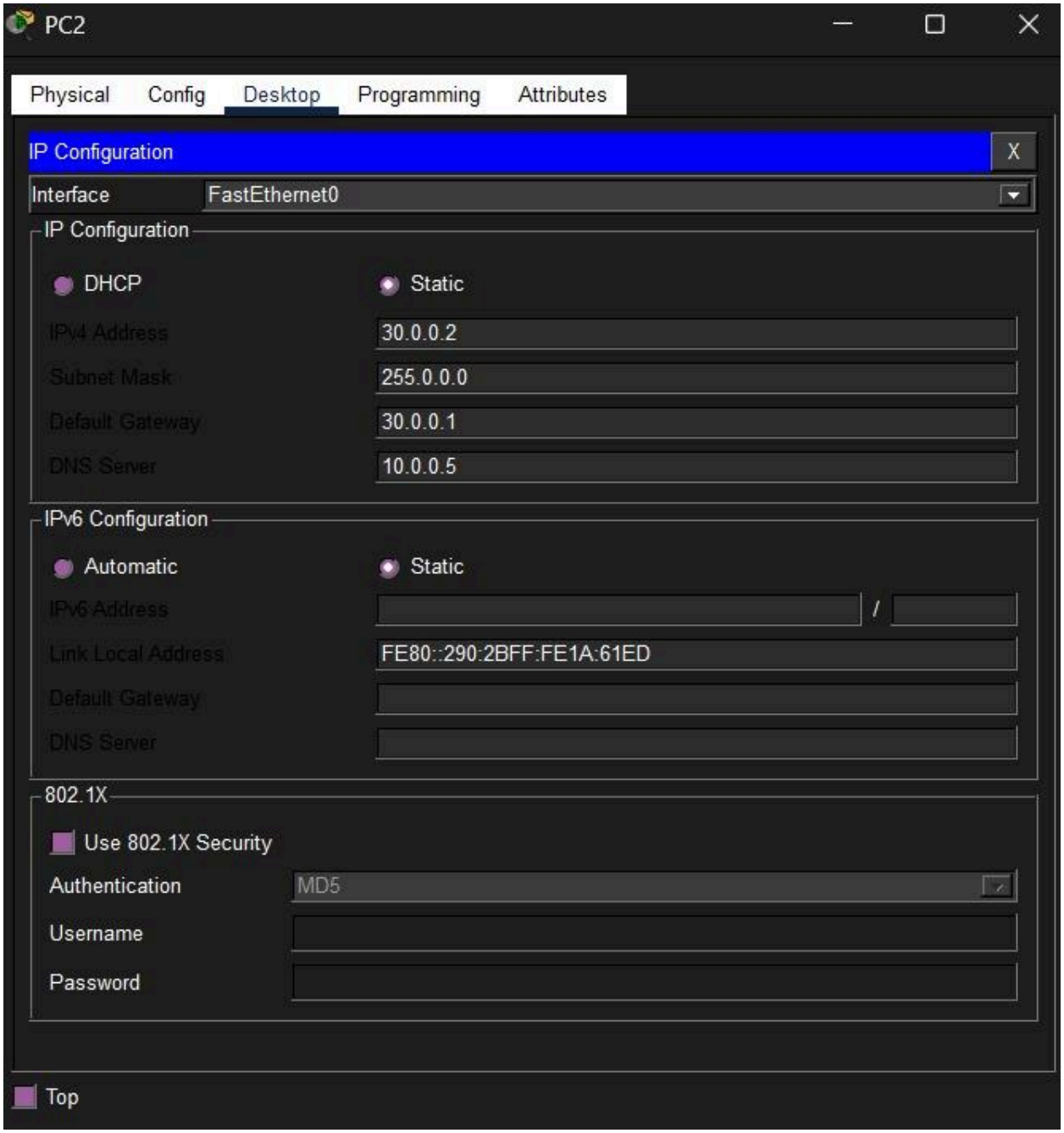
PC0


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
10.0.0.1
DNS Server
10.0.0.5
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::260:3EFF:FE49:CCC2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

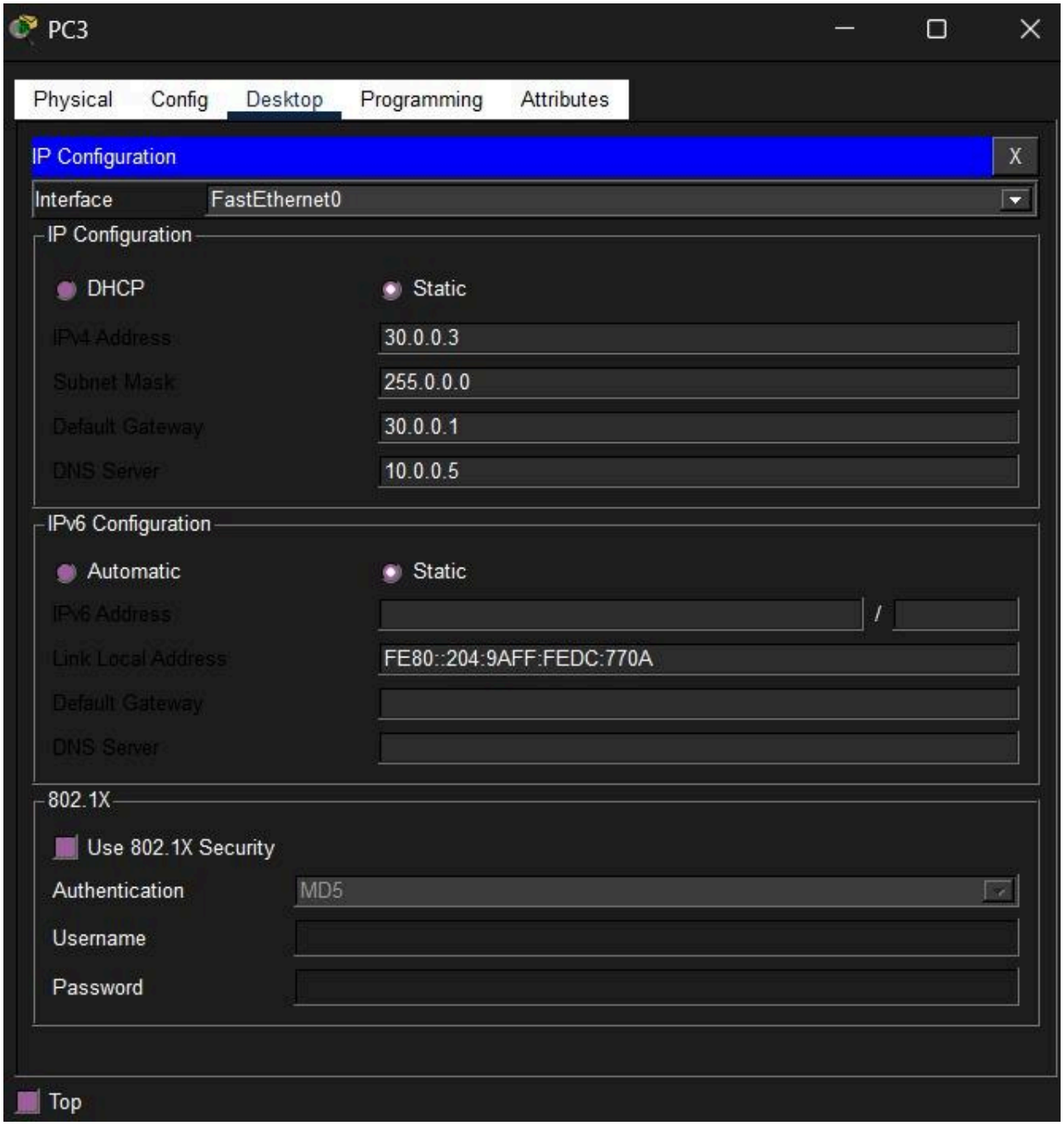
PC1


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.3
Subnet Mask
255.0.0.0
Default Gateway
10.0.0.1
DNS Server
10.0.0.5
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::20D:BDFF:FE1C:9C80
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

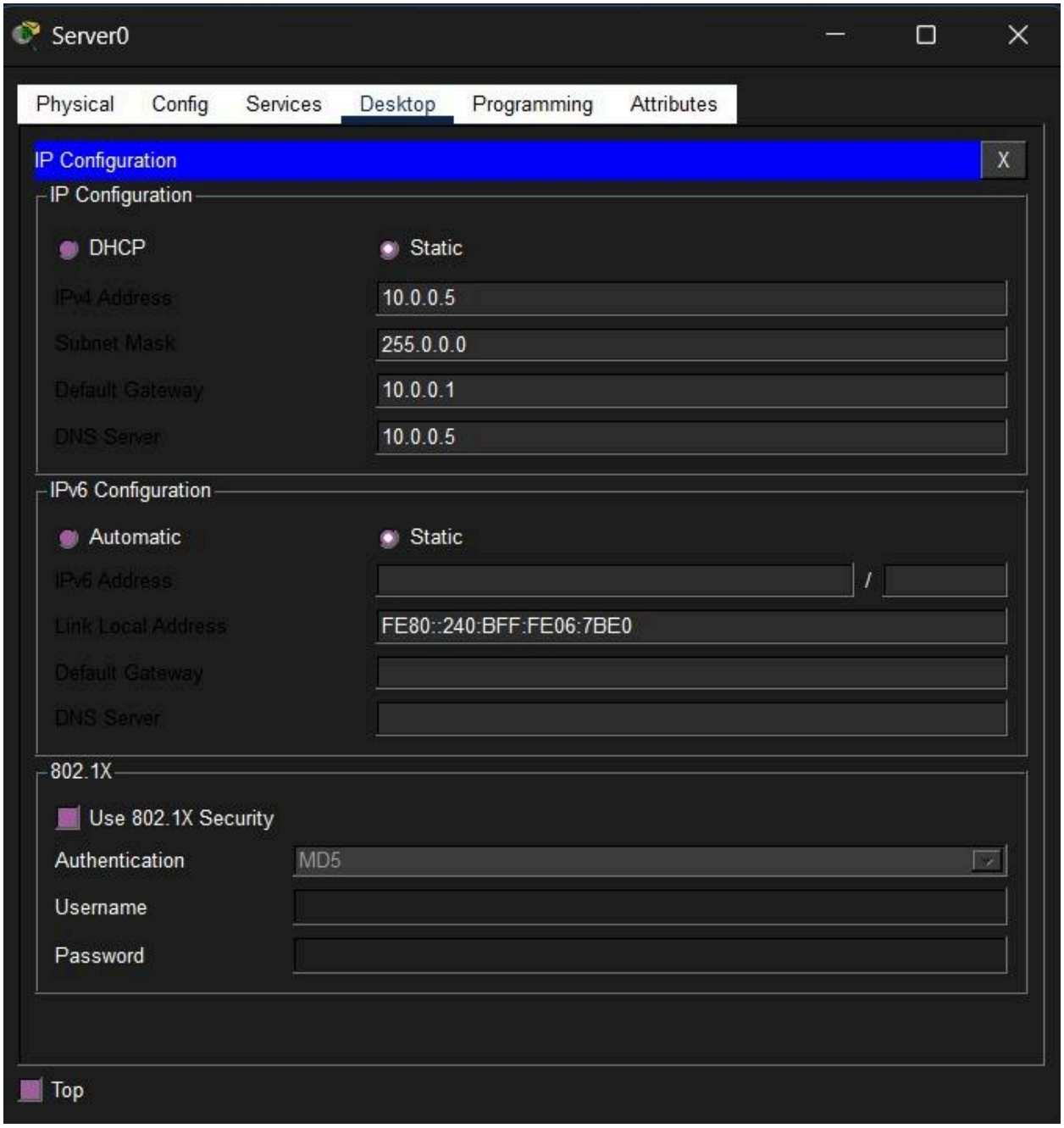
PC2

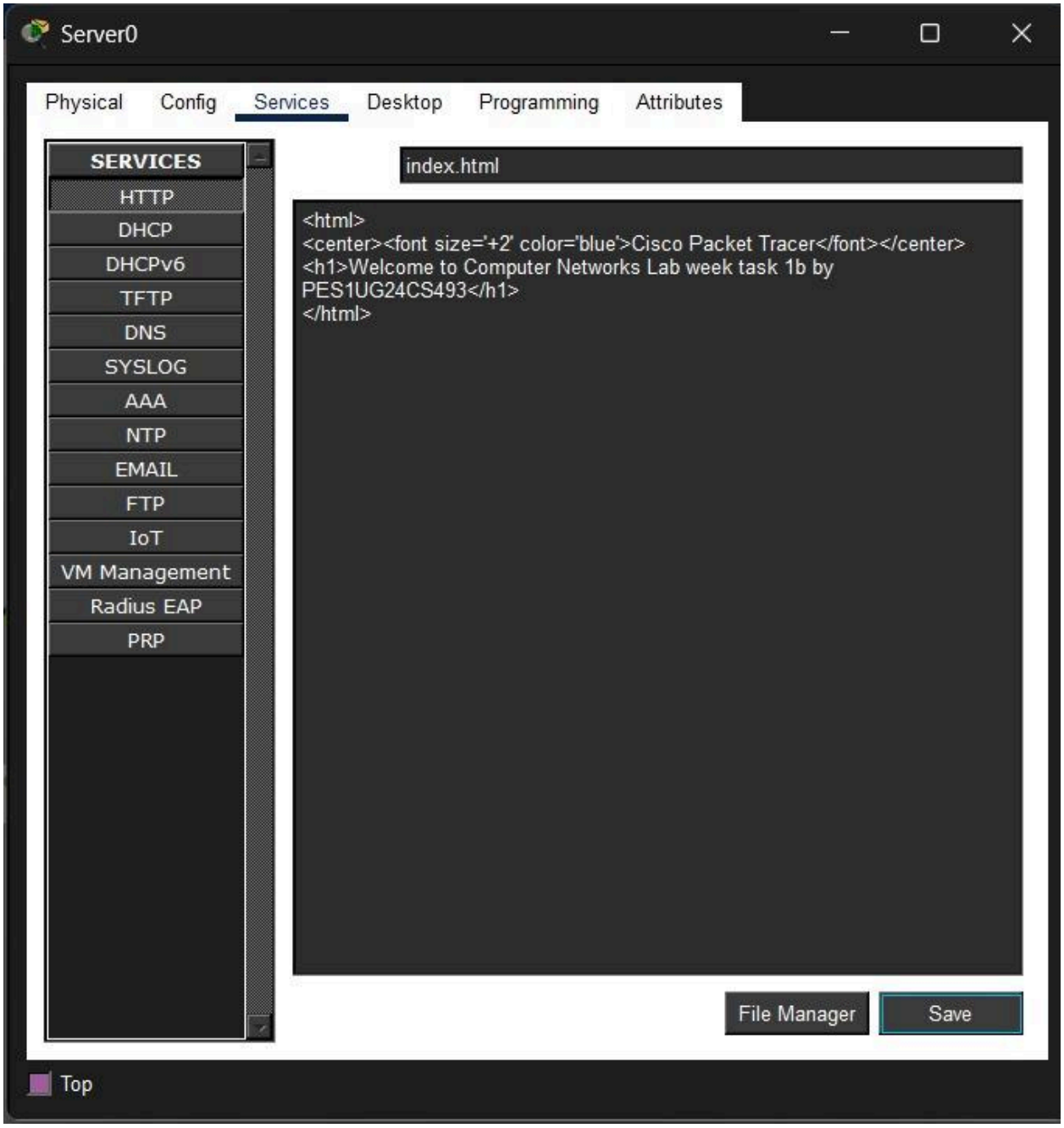

PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
30.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
30.0.0.1
DNS Server
10.0.0.5
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::290:2BFF:FE1A:61ED
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

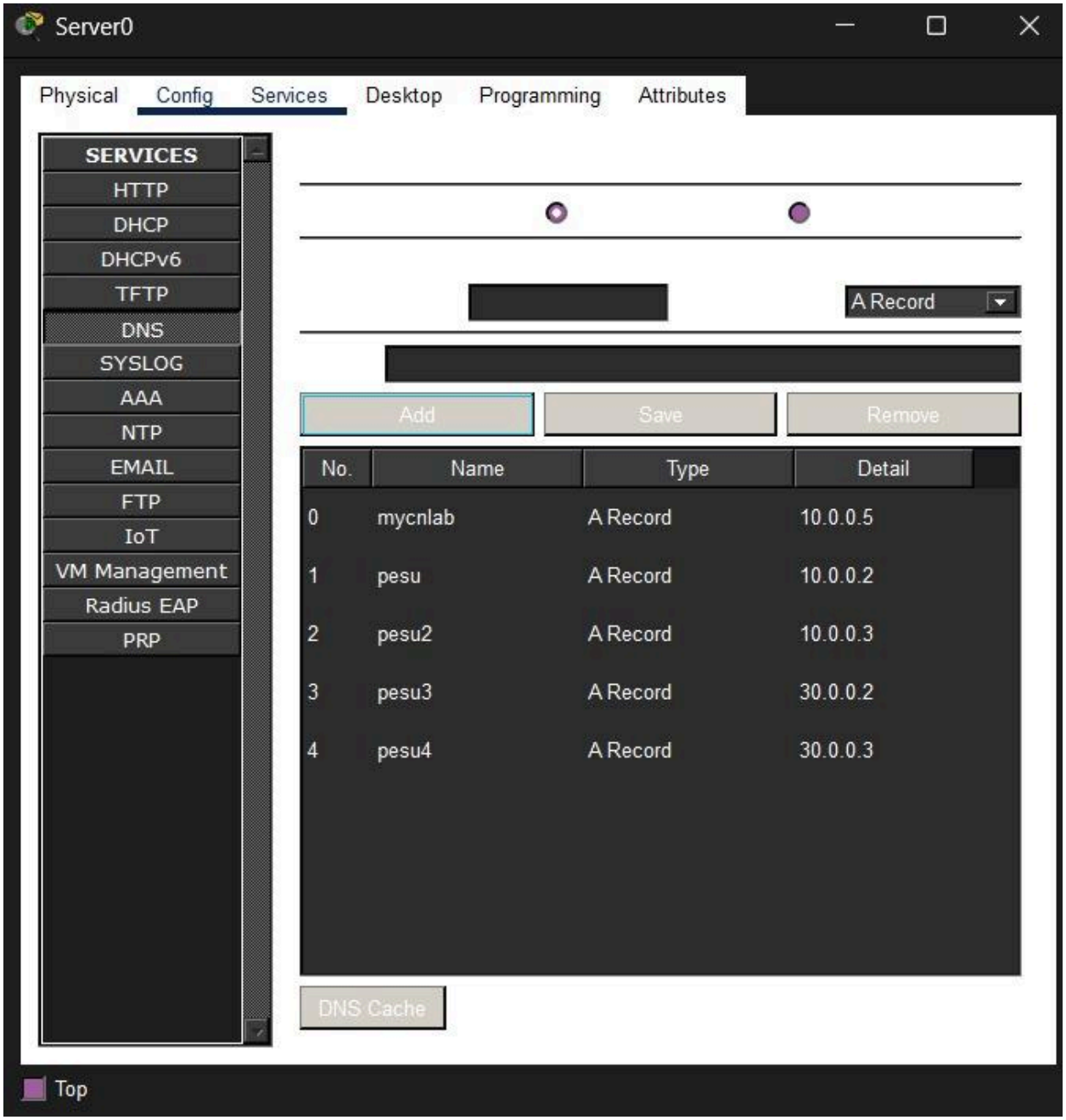
PC3


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
30.0.0.3
Subnet Mask
255.0.0.0
Default Gateway
30.0.0.1
DNS Server
10.0.0.5
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::204:9AFF:FEDC:770A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

SERVER


Server0
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.5
Subnet Mask
255.0.0.0
Default Gateway
10.0.0.1
DNS Server
10.0.0.5
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::240:BFF:FE06:7BE0
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
index.html
<html>
Cisco Packet Tracer
<h1>Welcome to Computer Networks Lab week task 1b by
PES1UG24CS493</h1>
</html>
File Manager
Save
Top


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
A Record
Add
Save
Remove
No.
Name
Type
Detail
0 mycnlab A Record 10.0.0.5
1 pesu A Record 10.0.0.2
2 pesu2 A Record 10.0.0.3
3 pesu3 A Record 30.0.0.2
4 pesu4 A Record 30.0.0.3
DNS Cache
Top